

Modeling Patent Challenges and Settlements in Pharmaceuticals

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Abstract

Generics can file Paragraph IV (PIV) patent challenges to work around a branded drug's patent and enter the market. Rather than pursuing litigation to completion, brand and generic manufacturers often reach a settlement that delays generic entry in return for payment, with potentially anticompetitive effects. I develop and calibrate a structural model of PIV challenges, settlement negotiations over launch date and payments, and litigation. I use the model to conduct policy counterfactuals involving Hatch-Waxman exclusivity length and assignment, no-authorized-generic commitments, accelerator clauses, and increased antitrust scrutiny of cash transfers.

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1 Introduction

Pharmaceuticals are a key driver of health improvements across the world; however, high pharmaceutical prices reduce access for patients until patent expiration. An important way of bringing down high prices is through generics filing a “Paragraph IV certification” to invalidate or work around a branded drug’s weak patent, and thus launch their product before patent expiration. However, rather than engaging in litigation, brand manufacturer and patent-challenging generic(s) often strike a settlement. This sees the generic drop the patent lawsuit and settle with the brand by agreeing to enter with a delay rather than try to invalidate the patent. In return, the brand pays the generic (via cash, exchanges, commitments to reduce competition upon generic entry, etc.). This benefits both the brand and the generic – the brand earns monopoly profit for several more periods, while the generic gets paid without having to incur the cost or uncertainty of a litigation loss. But consumers may lose, as they may have to pay high prices for longer.

Such settlements are highly controversial due to their potential anticompetitive effects, and have led to numerous lawsuits.¹ Various policies have been suggested that affect or ban settlement negotiations and clauses. What might be the effect of such policies, and can they improve consumer welfare? Answering these questions involves modeling strategic choices about filing patent challenges and negotiating over settlement terms by multiple market participants. The resulting model should also be rooted in data so that we may predict the effects of policy changes.

I build a structural model to study Paragraph IV filing decisions and settlement negotiations. The model is sequential and proceeds as follows. Potential generic manufacturers first decide whether to file a Paragraph IV (PIV) challenge against a brand drug.² The PIV filers are then assigned first-filer (FTF) or later-filer (LTF) status.³ The brand first bargains with the first filer(s) over launch date and cash transfer. If they settle, the brand bargains with the later filer(s) over launch dates and cash. If no mutually beneficial agreement exists, the relevant firms litigate. Then the market evolves, and product launch takes place for each agent based on their settlement/litigation launch-dates. Product-market profits accumulate from the date of realized launch, and are determined each period by the level of competition in the market. I represent antitrust scrutiny as a

¹Such settlements have been litigated many times in the US as well as in Europe. [Federal Trade Commission \(2010\)](#) estimated that settlements involving payment delayed generic entry by roughly 17 months relative to settlements without payment, at a cost to consumers of about \$3.5 billion per year. [Hemphill \(2006\)](#), [Hemphill and Lemley \(2011\)](#), and [Bulow \(2004\)](#) contain a detailed history of antitrust lawsuits and settlement terms. List of recent FTC actions against such settlements can be found here under “Press Releases”: [link](#).

²The “patent strength” of a drug reflects how likely it will survive litigation, and in reality many patents are invalidated in court. I also think of the patent strength of a drug as an aggregate of strengths of the various patents attached to the drug.

³First-filers file their PIVs earliest or on the first day such PIVs can be accepted by the FDA, while LTFs are those who file later. First-filer status can confer eligibility for the 180-day generic exclusivity period. This then leads to an asymmetry in bargaining status and terms compared to LTFs. In particular, LTFs have to settle for launch after the FTF has already launched.

cost of settlement that rises with cash payments, delay, and any additional gap between first- and later-filer entry. The model is solved by backward induction, and the settlement negotiations are modeled as Nash-bargaining.

An unfortunate issue with studying pharmaceutical settlements is lack of comprehensive data on settlements and terms. Settlements are generally confidential and their details are often not publicly revealed. For a handful of settlements, I can find their terms from antitrust cases filed in court and reported in the news. Thus, I try to reasonably calibrate the model parameters from prior literature and make use of the limited data in publicly-reported settlements. I conduct sensitivity checks by rerunning counterfactuals for values around the baseline calibration, and obtain similar results.

With the model set up and parameters calibrated, I solve the model for various counterfactuals to study their effects on equilibrium outcomes. I gauge the impact of policies on consumer welfare by studying earliest generic entry dates and total generics. An earlier generic entry date means that prices drop from monopoly levels earlier. More generics means greater competition, pushing down prices. Together these should lead to improvements in consumer welfare.

One criticism that has been raised about Hatch-Waxman exclusivity⁴ is that it is given to first-filers even if they settle for an anticompetitive pay-for-delay contract; thus, I study what happens if exclusivity is offered only to first-filers who invalidate the patent. In equilibrium the settlement launch dates are earlier, and total PIV filings remain the same in my simulations (the effect on entry incentives is small and ambiguous). Consumer welfare improves if total PIV filings stay the same – the first generic as well as the LTFs enter earlier, leading to an earlier decline from brand's monopoly prices.

Sometimes in lieu of payment the brand can commit to not marketing an Authorized Generic (AG),⁵ which is known as a no-AG commitment. When a no-AG commitment is included in the settlement contract, it leads to greater PIV filings (by one, in my simulations), and mostly unchanged settlement launch dates. Total cash transfer to generics declines, demonstrating how the no-AG commitment effectively substitutes for cash transfer in such reverse settlements. There is a likely negative impact on consumer welfare due to missing competition from the AG.

Some settlement contracts include an accelerator clause, which prevents FTFs who have already settled from being contractually held up when a LTF invalidates the patent and launches early. When an accelerator clause is included in the settlement contract, it leads to total PIV filings

⁴The Hatch-Waxman Act grants the first-filer 180 days of generic exclusivity, during which later-filers are blocked from launching. This temporary exclusivity is intended to reward the first-filer for incurring the costs and risks of challenging the patent.

⁵An Authorized Generic is chemically identical to the brand drug in terms of molecule structure and formulation, except it does not have the brand label attached. The AG therefore competes directly with the independent generics on prices, while the brand drug sets high prices. See [Alam and Conti \(2026\)](#) for details.

being broadly similar (though with the potential of being lower), settlement launch dates staying quite similar, and cash transfer to LTFs declining. Consumer welfare is generally unaffected.

Over time, the antitrust authorities have ramped up scrutiny of reverse settlements involving large cash payments, which in turn disincentivizes such reverse settlements. I study what happens when antitrust scrutiny of cash transfer is raised. Greater antitrust scrutiny of cash transfer leads to lower cash transfers, earlier settlement launch dates, and usually a decline in PIV filings. The net effect on consumer welfare is ambiguous – the first generic enters earlier, but the number of generic competitors may be weakly fewer.

There have been concerns that pay-for-delay deals incentivize drug manufacturers to focus on developing weak patents instead of strong patents, as weak patent monopolies can now be protected by such settlements. To see if this is likely, I study the brand's payoff from weak versus strong patents as cash settlements are banned. Banning cash in settlements has an ambiguous effect on the brand's payoff from weak versus strong patents.

I study what happens if the Hatch-Waxman exclusivity (HWE) length is increased; this has been suggested as a policy tool to incentivize more PIV certifications. As HWE length rises, FTF negotiates earlier settlement launch dates, LTF negotiates similar settlement launch dates, and total PIV filings stay similar with a chance of increasing. The effect on consumer welfare is therefore positive.

Finally, to illustrate PIV filing incentives within the model, I study how market outcomes vary if the same drug has different patent strengths. For stronger patents, there are fewer PIV filings, and both FTFs and LTFs choose later settlement dates.

Contributions: First, I contribute to the literature on pharmaceutical patent settlements. I am the first to set up and calibrate a structural model of reverse settlements and PIV filing decisions. In contrast to prior literature, my structural model allows me to cleanly work through the various interactions of multiple agents, choices, and clauses. It also allows me to solve for the equilibrium numerically rather than analytically, which expands the complexity I can incorporate into the model. The payoffs for each agent in my model accumulate via profits from subsequent evolution of the market.

I also contribute to the empirical literature on patent settlements and pay-for-delays. Unlike other papers, I construct a model that may be used to simulate and predict the effects of settlements and potential policy changes. Therefore, it may be used to aid antitrust cases on reverse settlements by using different parameter values to adjust to the setting.

While there is an extensive legal and theoretical literature studying reverse settlements, these papers use either intuition or theory models to work through policy implications. The limitation of such theory models is that they have to focus on a small subset of the issues to aid analytical

solutions, and so can't model many economic mechanisms at once. Without empirical values of key parameters, they also cannot quantify the magnitude of the mechanisms at play. Key references for understanding pay-for-delay settlements are [Hemphill \(2006\)](#) and [Hemphill and Lemley \(2011\)](#). Other useful references for the theory and legal literature on pay-for-delay include [Bokhari et al. \(2020\)](#), [Ding and Zhao \(2019\)](#), [Edlin et al. \(2013\)](#), [Edlin et al. \(2014\)](#), [Edlin et al. \(2015\)](#), [Scott Morton \(2013\)](#), [Shapiro \(2003\)](#), and [Bulow \(2004\)](#). For more, see [Gratz \(2012\)](#), [Mungan \(2013\)](#), and [Palikot and Pietola \(2023\)](#), and the references therein. On the empirical side, [Jacobo-Rubio et al. \(2020\)](#) and [Drake and McGuire \(2025\)](#) use stock price movements to study the effects of reverse settlements on brand payoffs and stakes. [Jacobo-Rubio et al. \(2020\)](#) also includes additional interesting findings on settlements and challenges from 1990s to 2000s. [Branstetter et al. \(2016\)](#) estimates the change in consumer and producer surplus from a PIV entry. [Ghili and Schmitt \(2016\)](#) uses Nash-bargaining to design a framework that may recover patent strength and put bounds on consumer welfare loss from pay-for-delay settlements, and contains interesting discussions on these issues.

Second, I contribute to the literature on generic entry. I am the first to model generic decisions to file Paragraph IV certifications while accounting for downstream settlement and litigation outcomes in a structural setting. [Alam and Conti \(2026\)](#) model generic entry and AG release decision without explicitly accounting for patent challenges. Some relevant papers studying generic entry using structural models include [Scott Morton \(1999\)](#), [Reiffen and Ward \(2005\)](#), and [Starc and Wollmann \(2023\)](#).

Third, I contribute to the literature on dynamics and bargaining in Empirical Industrial Organization. In contrast to most papers on Nash bargaining, my model accounts for dynamics in bargaining. The negotiation involves future launch dates alongside cash transfers. As a result, agreement and disagreement payoffs in bargaining are continuation values accounting for dynamic effects. [Igami and Uetake \(2020\)](#) estimate a stochastically alternating-move game of dynamic oligopoly to study merger decisions in the hard disk drive industry and solve the model by backward induction. [Lee and Fong \(2013\)](#) focus on the dynamics of network formation and bargaining.

Finally, I contribute to the literature on Law and Economics. Some papers building structural models to study settlements and bargaining in this literature are [Sieg \(2000\)](#), [Silveira \(2017\)](#), [Kong et al. \(2025\)](#), and [Merlo and Tang \(2019\)](#). I also contribute to the literature on antitrust and collusion; see [Asker and Nocke \(2021\)](#) for an extensive overview.

2 Institutional Background

The regulations governing Paragraph IV challenges shape PIV filing decisions and settlement terms. This section explains patent litigation, first-filer exclusivity, and the approval bottleneck,

then describes reverse settlements, antitrust scrutiny, and common settlement terms. Finally, I present evidence from prior literature on the frequency and terms of challenges and settlements. The elements discussed in this section will inform the model setup and calibration.

2.1 Pharmaceutical patents and Paragraph IV challenges

A drug manufacturer develops a product and patents it. While the patent is upheld, the innovator earns monopoly profits. Generics enter after patent expiration, or earlier through patent challenges and settlements. They market a bioequivalent product, thereby increasing competition and lowering prices. This paper focuses on generics' patent challenges and settlements that can lead to earlier entry. I refer to the original product as the "brand drug", its manufacturer as the "brand", and both independent generics and authorized generics (AGs) as "nonbrand" products.

Paragraph IV Certification: The generic maker submits an Abbreviated New Drug Application (ANDA) to the FDA for approval. To gain FDA approval, generic makers must prove to the FDA that they can manufacture products that are bioequivalent to the brand drug, at scale for commercial distribution, and according to Current Good Manufacturing Practices.

The ANDA must make a certification, of which the most relevant for this paper is Paragraph IV certification (henceforth PIV). A Paragraph IV certification asserts that the relevant patent is invalid, unenforceable, or will not be infringed by the proposed generic product.⁶

When filing a PIV application with the FDA, the generic entrant must also notify the brand manufacturer. If the brand files an infringement suit within 45 days, the FDA approval is generally stayed for up to 30 months while the litigation proceeds.

Probabilistic patents: Patents are "probabilistic", i.e. an issued patent is not a guarantee of monopoly rights or exclusion of rivals from the market. For instance, patents litigated in court are sometimes not upheld, generic manufacturers may also be able to design around a patent or create a bioequivalent product that does not infringe the patent, etc. [Lemley and Shapiro \(2005\)](#) state that of all patents litigated to a final determination, 46 percent are deemed invalid.⁷ For pharmaceuticals specifically, [Hemphill and Sampat \(2013\)](#) find that among challenges litigated to judgment, brands won on 92 percent of primary-patent challenges, but only 32 percent of secondary-patent

⁶A Paragraph IV certification may use different routes to argue the generic manufacturer can market the product despite the patent. For example, it may claim that the underlying molecule patent is invalid, that a manufacturing-process patent does not cover its product, that a listed patent is invalid or unenforceable, etc.

⁷That said, they also note that only 1.5 percent of all patents are ever litigated in the first place, so appropriate selection incentives apply. See [Farrell and Shapiro \(2008\)](#) and [Scott Morton \(2013\)](#) for more discussions on this topic.

challenges.⁸

Hatch-Waxman exclusivity: There is a free-riding incentive for filing PIV challenges – litigating a patent is costly, and a generic that wins removes this entry barrier for all other generics. Thus each generic hopes one of the others litigates and remove the patent. Without some reward for the challenger, a PIV challenge may not materialize even when facing a weak patent.

The Hatch-Waxman Act solves this by awarding the first Paragraph IV filer 180 days of exclusivity: for six months after it begins marketing, the FDA will not approve a competing ANDA for that drug. The first-filer therefore faces only the brand during the window, and can price far above the level that would prevail with several generics present.

Only the first filer can earn the 180-day exclusivity. Note that there can be multiple first applicants if they file on the same day, in which case all these FTFs share the exclusivity. I refer to these first applicants as first-filer (FTF) firms and to later Paragraph IV applicants as later-filer (LTF) firms. As a result, instead of free-riding, there is now a race to be the first-filer between all PIV filers.

Note that the Hatch-Waxman exclusivity does not prohibit the brand from marketing an Authorized Generic (AG) under its NDA, so an AG can compete with the FTF during the 180-day period. This increases the level of competition during the exclusivity period, and reduces the FTF's profits. As I explain later, withholding AG is therefore something the brand can offer the FTF during settlement negotiations.

For a detailed overview of the Hatch-Waxman Act and its implications for generic entry and Paragraph IV challenges, see [Hemphill and Sampat \(2025\)](#).

2.2 Reverse settlements

Reverse settlement: Instead of engaging in litigation with PIV filers, the brand can negotiate a settlement that avoids litigation. One common form of settlement is where the PIV filers agree to stay out until an agreed date, and in return the brand pays them in cash or other method of value transfer. Under these settlement terms, the PIV filer enters the market later than if litigation had invalidated the patent, resulting in longer periods of monopoly prices. Such settlements are commonly known as “reverse settlements” or “pay-for-delay”.⁹

⁸How can such “weak” patents make it through the patent office? Unfortunately, patent examination by the Patent and Trademark Office (PTO) is not foolproof; examiners spend approximately 18 hours per application, so many patents are later found to be invalid or unenforceable when litigated in court. [Lemley and Shapiro \(2005\)](#) explore additional issues behind this.

⁹In these settlements the patent holder pays the infringer, which is the reverse of what a normal infringement settlement looks like (where the infringer would pay the patent holder), hence the name “reverse settlements”.

There may be procompetitive or anticompetitive reasons behind such settlements. Procompetitive reasons are that settlement allows both parties to avoid litigation costs, and that the delay induced by settlement reflects the underlying patent strength. Anticompetitive reasons are that it allows the brand and PIV filers to redistribute surplus from the consumer to themselves.¹⁰ [Federal Trade Commission \(2010\)](#) estimated that settlements involving payment delayed generic entry by roughly 17 months relative to settlements without payment, at a cost to consumers of about \$3.5 billion per year.

Approval bottleneck: The FTF's 180-day clock does not start when the PIV is filed; rather, it starts when the first filer launches the product. If the FTF settles and stays off the market, the clock never starts, and the FDA cannot approve any LTFs. Settling with just the FTFs blocks the remaining generic queue.¹¹

The “approval bottleneck” refers to the fact that a later filer cannot launch until the first filer has launched and completed its Hatch-Waxman exclusivity period. If a LTF wants to launch in the absence of the FTF, its only real option is to make the FTF trigger or forfeit its exclusivity. The process of doing so involves continued litigation that is both very costly and highly uncertain. For detailed discussions on the approval bottleneck, see [Patel \(2009\)](#), [Hemphill and Lemley \(2011\)](#), and [Wiele et al. \(2021\)](#).

Even without the approval bottleneck, settlement with the FTF can worsen chances of early entry. The FTF is generally the applicant closest to launch and has the strongest incentive to litigate because of its eligibility for 180-day exclusivity. There is also concern that the FTF is the strongest candidate with the highest likelihood of winning litigation, so settling with the FTF removes the best chance of invalidating the patent. Thus, the removal of the FTF's challenge is economically consequential even when LTFs remain able to pursue approval ([Hemphill \(2006\)](#)).

Accelerator/Most-Favored-Entry clauses: An FTF that settles generally cannot enter before its settlement launch date. If an LTF unexpectedly obtains approval to enter earlier (usually through litigation), an accelerator clause would allow the FTF to enter earlier as well. The clause therefore prevents the FTF from being contractually held up when entry by a rival is imminent. Note that

¹⁰Why are such anticompetitive settlement deals stable and rational? Generic entry results in the brand losing most of its profits, the generic entrant receiving a small positive return, and consumers gaining from significantly lower prices. Under a pay-for-delay settlement, the brand can offer the generic in transfers more than the generic could earn by entering, and still retain a fraction of the monopoly profits. Both firms are better off than under generic entry, and the consumers lose as price competition is delayed.

¹¹In the past Congress has attempted to address this via the Medicare Modernization Act (MMA) of 2003. The MMA replaced “indefinite parking” with a set of forfeiture events that cause the FTF to trigger or forfeit its exclusivity, allowing LTFs to enter immediately or very soon. Unfortunately, the reform narrowed the bottleneck but did not close it. See [Patel \(2009\)](#), [Hemphill and Lemley \(2011\)](#), and [Wiele et al. \(2021\)](#) for discussions on this.

while the accelerator clause benefits the FTFs, it harms the LTFs.¹²

2.3 Antitrust enforcement and disguising transfers

Antitrust authorities have long recognized the potential harm of pay-for-delay settlements (e.g. [Federal Trade Commission \(2010\)](#)). Yet, antitrust challenges throughout the 2000s had little success due to negative rulings from courts.

The key reversal came in 2013 with Supreme Court's *Actavis* decision ([Edlin et al. \(2013\)](#), [Scott Morton \(2013\)](#)). This ruled that pay-for-delays should be judged under the rule of reason, rather than being per se legal or illegal. The court instructed that, rather than adjudicating the patent itself, courts should ask whether the reverse payment is large and unexplained. The brand manufacturer may pay for litigation costs plus the fair value of any goods or services the PIV filer actually supplies, but payment beyond that threshold may be purely to extract delay.

The *Actavis* ruling has had mixed results. On the one hand, *Actavis* made large, unexplained cash transfers legally dangerous, and as a result they have played a smaller role in the settlements since. On the other hand, settlements now increasingly look to transfer value to PIV filers through indirect methods where the value transfer is hard to quantify. See [Feldman and Frondorf \(2017\)](#) for a detailed discussion on such tactics.

There are several ways of disguising a cash transfer:

1. Obfuscation through side-deals: The brand manufacturer can include additional items in the settlement and obfuscate the use of the cash. One way is for the side-deals to include items like license exchanges which are valuable for the receiving filer but difficult to quantify in monetary terms. Another way is to receive items from the PIV filer and claim that a reverse payment is for compensation of those items. Furthermore, because avoided litigation cost is an admissible explanation, parties can claim the cash transfer to be reimbursement for litigation costs, and keep it low enough to avoid scrutiny.
2. No-AG commitment: The brand can commit to not marketing an Authorized Generic (AG), increasing the FTF's expected generic profits. This is a non-cash transfer, making it harder to quantify its monetary value and to litigate against in court. That said, courts and the FTC have increasingly treated no-AG commitments as a potential reverse payment.

¹²More generally, the accelerator clause is triggered by events signifying earlier-than-expected rival entry or competition. The FTC MMA report lists five common events that commonly trigger acceleration. These are (paraphrased from reports): the sale of a generic version of the product by another company; a final court decision that the patents are invalid, unenforceable, or not infringed; the licensing of a third party by the brand at an earlier launch date; a fall in brand sales below a specified threshold; and approval by the FDA of another product containing the same active ingredient.

2.4 Empirical patterns

Paragraph IV challenges have become increasingly common in the decades following the Hatch-Waxman Act.¹³ Yet, a drug’s effective market life – defined as the time from brand approval to first generic entry – has remained between 12 to 14 years since 2000, despite all the changes since. See [Hemphill and Sampat \(2025\)](#) for a discussion on the growth and composition of pharmaceutical patent challenges.

Conditional on facing any Paragraph IV challenges, how many distinct challengers does a drug typically face? The most relevant evidence comes from [Grabowski et al. \(2021\)](#). They state: “The average number of Paragraph IV challengers per NME (new molecular entity) varies by year of first generic entry, ranging between 2.8 (2005) and 8.6 (2018). The number of unique challengers for an NME ranged between one and 25, averaging 5.5 for the years 2004-19.”¹⁴

Since 2010, the number of settlements between brand and PIV filers ranged between 110-250 per year. A smaller fraction of this constitutes pay-for-delay settlements. For instance, in FY2021 the FTC reported 199 final settlements. Of these 33 were flagged for explicit compensation from brand to generic and restriction on generic’s ability to market the drug. (The compensation is almost always stated as for litigation fees.) 5 settlements include “possible compensation” because it is unclear if their provisions serve as compensation; such provisions may include not using a third-party to launch AG, declining royalty structure, quantity restriction on settling generic, etc. In contrast, in FY2010 there were 113 final settlements, of which 31 were classified by the FTC as “potential pay-for-delay”. In FY2009, there were 68 final settlements, of which 19 were classified as potential pay-for-delay.

To get a sense of the level of cash-transfer and delays negotiated in pay-for-delay settlements, I am able to find 14 settlements with publicly disclosed cash transfers. These have a median cash-transfer per generic of \$61m, with 10th and 90th percentiles of \$12m and \$216m. They have a median delay of 6.28 years ($\approx$ 25 quarters) with a minimum and maximum of 1.37 years and 9.42 years respectively. See Appendix B for the statistics and discussion on the collected terms.

Recall that no-AG commitments and accelerator clauses are sometimes part of the settlement terms, but how common are they? Accelerator clauses have recently been extremely common in patent settlements reported to the FTC. For instance, in FY2021 the FTC received 199 final settlements, of which 166 contained acceleration provisions. No-AG settlements became very

¹³[Hemphill and Sampat \(2025\)](#) state: “About 21 percent of drugs in the 1985-1987 approval cohort have one or more patents challenged by generics; this figure rises to 55 percent by 2000-2002. Nearly 80 percent of drugs approved between 2005 and 2010 face challenges, though the share declines in later years.”

¹⁴I can quantify the number of FTFs per New Drug Application (NDA) using data from the FDA. The FDA publishes a list of drug products for which a substantially complete Paragraph-IV certification has been submitted, and lists the number of first-filers in such cases. (It does not list the total number of PIV certifications per drug product). For the most updated list at the time of this writing (July 2026), there are 1623 NDAs with listed PIV certifications and FTFs. Of these, 85% of the NDAs have 1 FTF, the 95th percentile of the distribution is 3, and 98th percentile is 4.

frequent in the early 2010s, with about 10-20 per year. They have become rarer since, with about 1-4 per year between 2018-2021.

3 Model

I build a tractable structural model of Paragraph IV (PIV) challenges and settlement negotiations between the brand manufacturer and generic entrants. The game proceeds as follows:

3.1 Model preliminaries

Environment and timing: A brand manufacturer holds a patented drug and faces ex-ante identical potential generic entrants. Time is discrete, $t = 1, 2, \dots, \bar{T}$, with one period (equal to one quarter) and discount factor β . The patent expires at date $T < \bar{T}$.

The game has four stages:

1. Entry through PIV challenge: Potential entrants move sequentially. Each chooses to enter with a Paragraph IV (PIV) challenge at sunk cost κ_{piv} or stay out. Let K denote the number of PIV filers.

Number of postpatent generics P (i.e. non-PIV filers who enter after patent expiration/invalidation) is set exogenously. Let $N = K + P$ be the total number of generic entrants.

2. Nature assigns first-to-file (FTF) status: Each of the K filers becomes a first filer independently with probability $1/K$.¹⁵ Let M denote the realized number of FTFs, so $M \sim \text{Bin}(K, 1/K)$; the remaining $K - M$ filers are later filers (LTFs). The realized market structure is

$$\omega = (F, L, P), \quad F = \text{total FTFs}, \quad L = \text{total LTFs}, \quad P = N - K.$$

3. Settlement stage 1 (S1): The brand and the F FTFs negotiate a symmetric settlement contract (t_{s1}, C_{s1}) – a launch date of t_{s1} for the first filers and a lump-sum cash transfer C_{s1} paid to each FTF. If no mutually-beneficial contract exists, the FTFs litigate and the game ends with litigation.
4. Settlement stage 2 (S2): If the FTFs settled, the brand and the L LTFs negotiate a symmetric contract (t_{s2}, C_{s2}) – a launch date of t_{s2} for the later filers and a lump-sum cash transfer C_{s2} paid to each LTF, with t_{s2} constrained to lie at least the Hatch–Waxman exclusivity window Δ_{hwe} after t_{s1} . If no acceptable contract exists, the LTFs litigate.

¹⁵This indirectly reflects the “race-to-file”. As there are more PIV filers, the chances of any filer winning the FTF race declines.

5. Entry and profits realized: Products are launched and profits are realized based on negotiation outcomes.

I now explain the timing convention in this paper. The entry decision is taken first, and the two settlement stages follow t_e periods later, with no discounting between S1 and S2. I normalize the settlement negotiation date to $t = 1$ and index the profit timeline $t = 1, \dots, \bar{T}$ from there, so every date in the model – patent expiration T , the invalidation dates T_f^{inv}, T_ℓ^{inv} , and the negotiated settlement dates t_{s1}, t_{s2} – is measured relative to that. The entry decision therefore sits t_e periods before $t = 1$, where the sunk entry cost is paid.

There are two assumptions I make for simplicity:

1. Symmetric bargaining outcomes within a settlement stage: At each settlement stage, the brand bargains bilaterally with each generic firm participating in that stage. I impose symmetry across these bilateral bargains: each produces the same launch date and per-firm cash transfer, and either all bargains result in settlement or all result in litigation.
2. Comprehensive litigation following Stage-1 breakdown: If the brand and the FTFs cannot reach a mutually acceptable settlement, the settlement sequence ends. All Paragraph IV challenges are then resolved through litigation. In particular, the model does not allow the brand to litigate against the FTFs and subsequently negotiate a settlement with the LTFs.

These assumptions are discussed in detail in Section 4.

Payoff primitives: Per-period profits depend on the level of competition in the market. Specifically, they depend on the number of non-brand competitors n and presence of Authorized Generic $a \in \{0, 1\}$:

- $\pi^g(n, a)$: per-period profit of one generic firm when n non-brand products (including the AG, if present) are on the market;
- $\pi^B(n, a)$: per-period profit of the brand, inclusive of AG profit when $a = 1$.

Note that generic profits are symmetric across roles: FTFs and LTFs each earn $\pi^g(n, a)$ once they are in the market.

Timelines: A timeline records the date at which each generic cohort will reach the market, together with an indicator saying whether an authorized generic launches at all:

$$\tau = (t_f, t_\ell, t_p, a_\tau)$$

Here t_f , t_ℓ and t_p are the launch dates of the FTF, LTF and postpatent cohorts, and a_τ indicates whether an AG launches. A cohort that is empty at ω is assigned launch date ∞ .¹⁶ A timeline is determined partly by the negotiation outcomes in the settlement stages.

A timeline τ together with ω induces a deterministic path of market structure evolution, i.e. for each date t , the number of non-brand products on the market and whether an AG is present. For each date $t = 1, \dots, \bar{T}$, the number of non-brand products on the market and the presence of an AG are given by:

$$n(t, \tau, \omega) = F \mathbf{1}\{t \geq t_f\} + L \mathbf{1}\{t \geq t_\ell\} + P \mathbf{1}\{t \geq t_p\} + a_\tau \mathbf{1}\{t \geq t_a\}$$

$$a(t, \tau) = a_\tau \mathbf{1}\{t \geq t_a\}$$

where t_a is the AG's launch date. I assume that the timing of the AG launch is always the earliest generic launch date.

Discounted profit streams: For a timeline τ , the discounted profit stream is:

$$\Pi^r(\tau, \omega) = \sum_{t=1}^{\bar{T}} \beta^{t-1} \mathbf{1}\{t \geq t_r\} \pi^g(n_t, a_t), \quad r \in \{f, \ell\}$$

$$\Pi^B(\tau, \omega) = \sum_{t=1}^{\bar{T}} \beta^{t-1} \pi^B(n_t, a_t)$$

where f and ℓ refer to FTF and LTF respectively; $n_t = n(t, \tau, \omega)$ and $a_t \equiv a(t, \tau)$; Π^B is the brand manufacturer's profit stream (including brand and AG product). All profit streams are discounted to $t = 1$.

Timeline construction: An unchallenged or upheld patent expires at T .

1. Joint-settlement timeline: If both stages settle at (t_{s1}, t_{s2}) , then FTFs enter at t_{s1} and LTFs enter at t_{s2} .
2. FTF litigation: If the FTFs litigate,
 - (a) Patent upheld (prob. p_f): every generic cohort enters at expiration T .
 - (b) Patent invalidated (prob. $1 - p_f$): the FTF enters at the FTF invalidation date T_f^{inv} , all other generics enter Δ_{hwe} periods later.

¹⁶That is, $t_f = \infty$ if $F = 0$, $t_\ell = \infty$ if $L = 0$, and $t_p = \infty$ if $P = 0$.

3. LTF litigation given an FTF settlement: If the FTFs have settled at t_{s1} and the LTFs litigate,

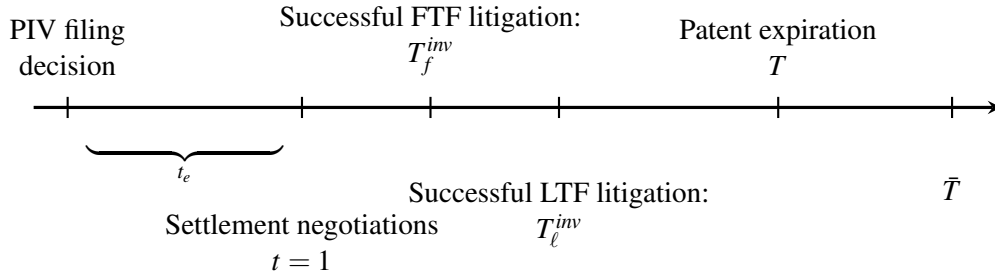
- (a) Patent upheld (prob. p_ℓ): FTFs enter at t_{s1} , LTFs enter at T .
- (b) Patent invalidated (prob. $1 - p_\ell$): FTFs enter at t_{s1} , LTFs enter at the LTF invalidation date T_ℓ^{inv} .

Postpatent generics enter when the patent is invalidated (potentially delayed by the Hatch-Waxman exclusivity length) or expires, whichever happens earlier. See Table 1 for exact details. Figure 1 includes a timeline of all non-settlement dates to aid visualization.

Table 1 lists every timeline the model constructs.¹⁷ The labels follow three rules:

1. When both settlement stages settle, $\tau = \tau^{sett}$.
2. When there are no Paragraph IV challenges, $\tau = \tau^{exp}$.
3. When a stage litigates, the superscript says which stage started the litigation (S1 or S2), and the subscript says whether that litigation left the patent upheld (U) or invalidated (I).

Figure 1: Timeline of Paragraph IV filing and non-settlement launch dates.



The variable p_f (and relatedly p_ℓ) captures the “patent strength” of the brand drug, i.e. how likely the patent will be upheld if litigated in court.¹⁸

AG regimes and Accelerator clauses: The model contains two contractual clauses. The first covers the potential inclusion of a no-AG commitment from the brand, and the second covers the potential inclusion of the accelerator clause. Neither clause is negotiated – the agents only bargain over an launch date and a cash transfer, (t_s, C_s) , not on the inclusion of the clauses themselves. The inclusion of a clause is set exogenously.

¹⁷Note that on the S2-litigation branch $a_\tau = a_{set}$, not a_{lit} : the FTFs have already settled, so their contract governs the AG whatever the LTFs do in court. In particular, under the no-AG regime $(a_{set}, a_{lit}) = (0, 1)$, an LTF suit does not trigger the AG — the no-AG clause in the FTF’s settlement continues to bind.

¹⁸In reality, each drug is covered by multiple patents. These patents could each be of differing strengths. Instead of modeling each patent and its expiration date separately, I think of the patent strength and relevant dates as reflecting the overall attributes of all patents associated with the drug.

Table 1: Notation for various constructed timelines.

Timeline	Arises when	(t_f, t_ℓ, t_p)	a_τ
τ^{sett}	FTF settles, and the LTF stage settles or does not exist	(t_{s1}, t_{s2}, T)	a_{set}
τ^{exp}	no challenge ($K = 0$)	(T, T, T)	0
τ_U^{s1}	FTF litigates, patent upheld	(T, T, T)	a_{lit}
τ_I^{s1}	FTF litigates, patent invalidated	$(T_f^{inv}, T_f^{inv} + \Delta_{hwe}, T_f^{inv} + \Delta_{hwe})$	a_{lit}
τ_U^{s2}	FTF settled, LTF litigates, patent upheld	(t_{s1}, T, T)	a_{set}
τ_I^{s2}	FTF settled, LTF litigates, patent invalidated	$(t_{s1}, T_\ell^{inv}, T_\ell^{inv})$	a_{set}

Notes: This table includes some parameters defined in later sections.

Here Δ_{hwe} is the Hatch–Waxman exclusivity window. The LTF-invalidation row τ_I^{s2} is stated for the baseline without the accelerator clause ($\mathbb{A} = 0$); the clause modifies it as described nearby.

1. The AG regime: The AG regime determines whether the brand would launch an AG in the event of a settlement or litigation by FTF. It is denoted by:

$$(a_{set}, a_{lit}) \in \{0, 1\} \times \{0, 1\}.$$

This is interpreted as:

- When S1 resolves in a settlement, AG release decision follows a_{set} .
- When S1 resolves in litigation, it follows a_{lit} .
- If S1 resolves in settlement and S2 in litigation, the AG regime is still a_{set} . The FTFs' contract binds regardless of what the LTFs do in court. However, the timing of the AG launch is always the earliest generic launch date.

I also assume for simplicity that when there are no PIV challenges, no AG launches.¹⁹

The possible AG regimes are:

- (a) AG never released: $(a_{set}, a_{lit}) = (0, 0)$. This is the baseline calibration.

¹⁹See [Alam and Conti \(2026\)](#) for AG release decisions after patent expiration.

(b) No-AG settlement: $(a_{set}, a_{lit}) = (0, 1)$. The AG is withheld when the FTFs settle, and launched when they litigate.

(c) Unconditional AG release: $(a_{set}, a_{lit}) = (1, 1)$.

2. Accelerator clause: The accelerator clause in FTF's settlement contract is denoted by $\mathbb{A} \in \{0, 1\}$.

This clause is activated when FTFs settle but LTFs litigate and successfully invalidate the patent. It protects FTFs against LTFs invalidating the patent and entering early while the FTFs are held to their settlement launch date. With it, the FTFs' entry is accelerated.

The accelerator clause thus modifies the LTF-invalidation timeline τ_I^{s2} from Table 1:

$$\mathbb{A} = 1 : \quad (T_\ell^{inv}, T_\ell^{inv} + \Delta_{hwe}, T_\ell^{inv} + \Delta_{hwe}).$$

Without the accelerator clause, the FTFs stay bound to their contracted date t_{s1} while the LTFs enter at T_ℓ^{inv} . With the accelerator clause, the FTFs re-anchor to T_ℓ^{inv} and the LTFs are held Δ_{hwe} behind them.

The antitrust scrutiny function: During settlement, the agents should take into account potential antitrust scrutiny of any settlement terms. For instance, a settlement with large cash transfers and long delays is likely to attract antitrust scrutiny from the FTC and DOJ. This then invites the possibility of antitrust litigation and associated costs and risks. I model this as a penalty that forces manufacturers to agree to terms with lower cash and delays than otherwise.

In addition, most reported pay-for-delay settlements and other public evidence see LTFs entering right after the expiration of the Hatch-Waxman exclusivity for FTFs. To enforce this, I impose the antitrust scrutiny to also be increasing in the gap between FTF and LTF settlement dates. This can be interpreted as antitrust scrutiny being engaged if LTFs settle significantly later than FTFs. See Appendix B for a discussion of the empirical evidence on this.

Settling parties bear antitrust penalty of Ψ_{s1} on the brand-FTF contract and Ψ_{s2} on the brand-LTF contract:

$$\begin{aligned} \Psi_{s1}(C_{s1}, t_{s1}, F) &= F \left[\psi_1 C_{s1} + \psi_2 C_{s1}^2 + \psi_3 E_{s1} + \psi_4 E_{s1}^2 \right], \\ \Psi_{s2}(C_{s2}, t_{s2}, t_{s1}, L) &= L \left[\psi_1 C_{s2} + \psi_2 C_{s2}^2 + \psi_3 E_{s2} + \psi_4 E_{s2}^2 + \psi_5 G + \psi_6 G^2 \right]. \end{aligned}$$

where:

1. Scrutiny over cash: C_{s1} and C_{s2} are the cash transfers in Stages S1 and S2 respectively.

2. Scrutiny over delay: E_{s1} and E_{s2} are the excesses of the settlement dates over their respective scrutiny-date benchmarks b_{s1} and b_{s2} .

$$E_s = \max\{0, t_s - b_s\}.$$

The scrutiny-date benchmark is each stage's expected launch date under litigation:

$$b_{s1} = p_f T + (1 - p_f) T_f^{inv}, \quad b_{s2} = p_\ell T + (1 - p_\ell) T_\ell^{inv}.$$

Scrutiny thus applies only to delay beyond what patent strength justifies. Below the benchmark the marginal penalty on delay is zero.²⁰

3. Scrutiny over gap: G is the gap between the FTF and LTF settlement dates beyond the statutory exclusivity window:

$$G(t_{s1}, t_{s2}) = \max\{0, t_{s2} - t_{s1} - \Delta_{hwe}\},$$

defined when both stages settle and $L \geq 1$.

The gap term is present in Ψ_{s2} but not in Ψ_{s1} . This means that it disciplines the stage-2 settlement date: it forces the LTF to settle close to the FTF's settlement date.

3.2 Settlement bargaining

Next I describe the model's two settlement negotiations. The brand first bargains with the FTFs over a launch date and payment, recognizing that any agreement will be followed by a negotiation with LTFs. If either negotiation fails, the relevant challengers litigate; failure at the first stage sends all challengers to litigation. The negotiations are modeled as Nash bargaining problems with the antitrust scrutiny penalty included in the payoffs. The subsection presents these negotiations in reverse order, beginning with LTFs.

3.2.1 Stage S2: the brand-LTF bargain

Fix $\omega = (F, L, P)$ with $L \geq 1$ and a signed S1 contract (t_{s1}, C_{s1}) . The brand and the L LTFs negotiate (t_{s2}, C_{s2}) with $t_{s2} \in \mathcal{T}_2(t_{s1}) = \{t_{s1} + \Delta_{hwe}, t_{s1} + \Delta_{hwe} + 1, \dots, T\}$ (i.e. the LTF cannot enter until FTFs have entered and the Hatch-Waxman exclusivity period has expired) and $C_{s2} \in [0, \infty)$.

²⁰Note that there are alternatives for setting b_{s1} and b_{s2} . For instance, it could be arbitrarily set to 4 or 8 quarters after the settlement negotiation date. In my simulations, I find that the current setup of expected launch date under litigation significantly reduces the chances of PIV filers choosing to litigate. I discuss the motives and tradeoffs in Section 4.

Agreement payoffs: Pre-transfer product-market values on the joint-settlement timeline:

$$\bar{V}_{s2}^B(t_{s1}, t_{s2}, \omega) = \Pi^B(\tau^{sett}, \omega), \quad \bar{V}_{s2}^\ell(t_{s1}, t_{s2}, \omega) = \Pi^\ell(\tau^{sett}, \omega).$$

If the brand and LTFs settle, the brand pays C_{s2} to each LTF, and they incur the antitrust scrutiny penalty Ψ_{s2} each.

On top of the stage-2 items, the brand's payoff carries its stage-1 bill – the FTF cash and the S1 antitrust penalty – pinned by the signed contract (t_{s1}, C_{s1}) and owed however the LTF stage resolves. Thus, the agreement payoffs are:

$$W_{s2}^B(t_{s2}, C_{s2}, t_{s1}, C_{s1}, \omega) = \underbrace{\bar{V}_{s2}^B - LC_{s2} - \Psi_{s2}(C_{s2}, t_{s2}, t_{s1}, L)}_{\tilde{W}_{s2}^B(t_{s2}, C_{s2}, t_{s1}, \omega)} - FC_{s1} - \Psi_{s1}(C_{s1}, t_{s1}, F),$$

$$W_{s2}^\ell(t_{s2}, C_{s2}, t_{s1}, \omega) = \bar{V}_{s2}^\ell + C_{s2} - \Psi_{s2}(C_{s2}, t_{s2}, t_{s1}, L).$$

The brace defines the stage-2 component $\tilde{W}_{s2}^B(t_{s2}, C_{s2}, t_{s1}, \omega)$; note that C_{s1} is not among its arguments. The LTF's payoff carries no stage-1 term.

Disagreement payoffs: The threat point is the LTF litigation lottery less litigation costs (ℓ_ℓ^B per LTF for the brand; ℓ^G for each LTF). The brand's stage-1 bill is owed on this branch too, as the S1 contract stands whether or not the LTFs litigate:²¹

$$\Omega_{s2}^B(t_{s1}, C_{s1}, \omega) = \underbrace{p_\ell \Pi^B(\tau_U^{s2}, \omega) + (1 - p_\ell) \Pi^B(\tau_I^{s2}, \omega) - L\ell_\ell^B}_{\tilde{\Omega}_{s2}^B(t_{s1}, \omega)} - FC_{s1} - \Psi_{s1}(C_{s1}, t_{s1}, F).$$

$$\Omega_{s2}^\ell(t_{s1}, \omega) = p_\ell \Pi^\ell(\tau_U^{s2}, \omega) + (1 - p_\ell) \Pi^\ell(\tau_I^{s2}, \omega) - \ell^G.$$

Gains from trade and Nash bargaining:

$$\text{GFT}_{s2}^B(t_{s2}, C_{s2}, t_{s1}, \omega) = W_{s2}^B(t_{s2}, C_{s2}, t_{s1}, C_{s1}, \omega) - \Omega_{s2}^B(t_{s1}, C_{s1}, \omega),$$

$$\text{GFT}_{s2}^\ell(t_{s2}, C_{s2}, t_{s1}, \omega) = W_{s2}^\ell(t_{s2}, C_{s2}, t_{s1}, \omega) - \Omega_{s2}^\ell(t_{s1}, \omega).$$

²¹Note that Ω_{s2}^ℓ depends on t_{s1} : without the accelerator clause, a later FTF settlement date leaves the LTF alone in the market for longer on the invalidation branch, so the LTF's threat point rises in t_{s1} ; with the clause it is flat.

Notice that GFT_{s2}^B is no longer a function of C_{s1} . This is because terms including C_{s1} cancel out when differencing the brand's agreement and disagreement payoffs. The stage-1 terms $-F C_{s1} - \Psi_{s1}(C_{s1}, t_{s1}, F)$ enter W_{s2}^B and Ω_{s2}^B identically: they are pinned by the signed S1 contract and paid whether the LTFs settle or litigate. They therefore cancel from the brand's gains from trade:

$$\begin{aligned}\text{GFT}_{s2}^B(t_{s2}, C_{s2}, t_{s1}, \omega) &= W_{s2}^B(t_{s2}, C_{s2}, t_{s1}, C_{s1}, \omega) - \Omega_{s2}^B(t_{s1}, C_{s1}, \omega) \\ &= \tilde{W}_{s2}^B(t_{s2}, C_{s2}, t_{s1}, \omega) - \tilde{\Omega}_{s2}^B(t_{s1}, \omega)\end{aligned}$$

These terms are the only place C_{s1} appears on either side, so GFT_{s2}^B — and with it the Nash product and the first-order condition — is not a function of C_{s1} .²²

The equilibrium S2 terms maximize the Nash product:

$$(t_{s2}^*, C_{s2}^*)(t_{s1}, \omega) = \arg \max_{t_{s2} \in \mathcal{T}_2(t_{s1}), C \in [0, \infty)} [\text{GFT}_{s2}^B(t_{s2}, C)]^\alpha [\text{GFT}_{s2}^\ell(t_{s2}, C)]^{1-\alpha}$$

subject to the constraints that $\text{GFT}_{s2}^B > 0$ and $\text{GFT}_{s2}^\ell > 0$.

The LTF response: Define the LTF response to each feasible FTF date:

$$\rho(t_{s1}, \omega) = \begin{cases} (\text{settle}; t_{s2}^*(t_{s1}, \omega), C_{s2}^*(t_{s1}, \omega)) & \text{if } (t_{s2}^*, C_{s2}^*) \text{ exist for } (t_{s1}, \omega), \\ \text{litigate} & \text{otherwise.} \end{cases}$$

3.2.2 Stage S1: the brand–FTF bargain

The brand and the F FTFs negotiate (t_{s1}, C_{s1}) with $t_{s1} \in \mathcal{T}_1 = \{T_f^{\text{inv}}, T_f^{\text{inv}} + 1, \dots, T\}$ and $C_{s1} \in [0, \infty)$, anticipating the downstream LTF response $\rho(\cdot, \omega)$.

Agreement payoffs: The brand's pre-transfer S1 value is its profit stream value net of the downstream LTF deal; the FTF's is its own profit stream on the induced path.

If $\rho(t_{s1}, \omega) = (\text{settle}, t_{s2}, C_{s2})$:

$$\bar{V}_{s1}^B(t_{s1}, \rho(t_{s1}, \omega), \omega) = \Pi^B(\tau^{\text{sett}}, \omega) - L C_{s2} - \Psi_{s2}(C_{s2}, t_{s2}, t_{s1}, L),$$

$$\bar{V}_{s1}^f(t_{s1}, \rho(t_{s1}, \omega), \omega) = \Pi^f(\tau^{\text{sett}}, \omega).$$

²²Importantly, this allows me to solve the model by backward induction. The stage-2 problem depends on the S1 contract only through t_{s1} and never through C_{s1} . This is what allows the solution method of Section 3.2.2 to compute the LTF response $\rho(t_{s1}, \omega)$ for every candidate FTF date before C_{s1} is determined; if C_{s1} entered GFT_{s2}^B , the two stages would instead have to be solved jointly.

If $\rho(t_{s1}, \omega) = \text{litigate}$:

$$\bar{V}_{s1}^B(t_{s1}, \rho(t_{s1}, \omega), \omega) = p_\ell \Pi^B(\tau_U^2, \omega) + (1 - p_\ell) \Pi^B(\tau_I^2, \omega) - L\ell_\ell^B,$$

$$\bar{V}_{s1}^f(t_{s1}, \rho(t_{s1}, \omega), \omega) = p_\ell \Pi^f(\tau_U^2, \omega) + (1 - p_\ell) \Pi^f(\tau_I^2, \omega).$$

Agreement payoffs:²³

$$W_{s1}^B(t_{s1}, C_{s1}, \rho(t_{s1}, \omega), \omega) = \bar{V}_{s1}^B - F C_{s1} - \Psi_{s1}(C_{s1}, t_{s1}, F),$$

$$W_{s1}^f(t_{s1}, C_{s1}, \rho(t_{s1}, \omega), \omega) = \bar{V}_{s1}^f + C_{s1} - \Psi_{s1}(C_{s1}, t_{s1}, F).$$

Disagreement payoffs: If the FTFs litigate, the game ends: all $K = F + L$ filers litigate at once, resulting in the patent being upheld with probability p_f and invalidated otherwise. The brand pays one suit's cost per filer (at the S1 per-suit price), and each filer pays its own cost:

$$\Omega_{s1}^B(\omega) = p_f \Pi^B(\tau_U^1, \omega) + (1 - p_f) \Pi^B(\tau_I^1, \omega) - (F + L) \ell_f^B,$$

$$\Omega_{s1}^f(\omega) = p_f \Pi^f(\tau_U^1, \omega) + (1 - p_f) \Pi^f(\tau_I^1, \omega) - \ell^G.$$

On this branch each LTF earns:

$$\Omega_{s1}^\ell(\omega) = p_f \Pi^\ell(\tau_U^1, \omega) + (1 - p_f) \Pi^\ell(\tau_I^1, \omega) - \ell^G$$

Gains from trade and Nash-bargaining: Gains from trade are

$$\text{GFT}_{s1}^B(t_{s1}, C_{s1}, \rho(t_{s1}, \omega), \omega) = W_{s1}^B(t_{s1}, C_{s1}, \rho(t_{s1}, \omega), \omega) - \Omega_{s1}^B(\omega),$$

$$\text{GFT}_{s1}^f(t_{s1}, C_{s1}, \rho(t_{s1}, \omega), \omega) = W_{s1}^f(t_{s1}, C_{s1}, \rho(t_{s1}, \omega), \omega) - \Omega_{s1}^f(\omega).$$

Stage 1 adds an element that stage 2 does not have – every candidate date accounts for the downstream LTF response $\rho(t_{s1}, \omega)$ that the date induces.

The equilibrium S1 terms maximize the Nash product:

$$(t_{s1}^*, C_{s1}^*)(\omega) = \arg \max_{t_{s1} \in \mathcal{T}_1, C \in [0, \infty)} [\text{GFT}_{s1}^B(t_{s1}, C)]^\alpha [\text{GFT}_{s1}^f(t_{s1}, C)]^{1-\alpha}$$

²³Substituting $\bar{V}_{s1}^B$ makes the brand's full bill explicit. When the LTFs settle, $W_{s1}^B = \Pi^B(\tau^{sett}) - F C_{s1} - \Psi_{s1} - L C_{s2} - \Psi_{s2}$; when they litigate, $W_{s1}^B = p_\ell \Pi^B(\tau_U^2) + (1 - p_\ell) \Pi^B(\tau_I^2) - F C_{s1} - \Psi_{s1} - L \ell_\ell^B$ (arguments suppressed). The brand pays its own stage-1 cash and penalty on both branches. What changes is the downstream item – the brand pays the LTFs' cash and stage-2 penalty if they settle, or pays litigation cost if they litigate.

subject to the constraints that $\text{GFT}_{s1}^B > 0$ and $\text{GFT}_{s1}^f > 0$.

If such a maximizer does not exist, the FTFs litigate and the game ends at the S1 disagreement point.

3.3 Entry and Equilibrium

Conditional payoffs at a fixed ω : Solving S2 then S1 at ω yields $\chi^f(\omega) \in \{\text{settle}, \text{litigate}\}$, $\chi^\ell(\omega) \in \{\text{settle}, \text{litigate}\}$ (defined when $\chi^f = \text{settle}$ and $L \geq 1$),²⁴ terms $(t_{s1}^*, C_{s1}^*, t_{s2}^*, C_{s2}^*)(\omega)$, and conditional values for each role:

$$V^B(\omega) = \begin{cases} W_{s1}^B(t_{s1}^*, C_{s1}^*, \rho(t_{s1}^*, \omega), \omega) & \chi^f = \text{settle}, \\ \Omega_{s1}^B(\omega) & \chi^f = \text{litigate}, \end{cases}$$

$$V^f(\omega) = \begin{cases} W_{s1}^f(t_{s1}^*, C_{s1}^*, \rho(t_{s1}^*, \omega), \omega) & \chi^f = \text{settle}, \\ \Omega_{s1}^f(\omega) & \chi^f = \text{litigate}, \end{cases}$$

$$V^\ell(\omega) = \begin{cases} W_{s2}^\ell(t_{s2}^*, C_{s2}^*, t_{s1}^*, \omega) & \chi^f = \text{settle}, \chi^\ell = \text{settle}, \\ \Omega_{s2}^\ell(t_{s1}^*, \omega) & \chi^f = \text{settle}, \chi^\ell = \text{litigate}, \\ \Omega_{s1}^\ell(\omega) & \chi^f = \text{litigate}, \end{cases}$$

Note that $V^B(\omega)$ subtracts every payment the brand makes on the realized path to FTFs and LTFs for settlement and litigation outcomes.²⁵

The FTF race and expected role values: Consider $K \geq 1$ filers and $N \geq K$ total entrants. Each filer becomes an FTF independently with probability $1/K$. Let M be the total realized FTFs; then the structure realized at the settlement stages is $\omega_M = (M, K - M, P)$ for $M = 0, 1, \dots, K$. When $M = 0$ all K filers are promoted to FTFs, so every object at ω_0 is evaluated at $\omega_K = (K, 0, P)$ instead. In particular $V^\ell(\omega_0) = V^f(\omega_K)$.

For a firm filing a PIV, with probability $1/K$ it is a FTF; the other $K - 1$ filers then contribute $M - 1 \sim \text{Bin}(K - 1, 1/K)$ additional FTFs. With probability $(K - 1)/K$ it is a LTF; the others contribute $M \sim \text{Bin}(K - 1, 1/K)$ FTFs. Writing $B_{n,q}(j) = \binom{n}{j} q^j (1 - q)^{n-j}$ with $q = 1/K$, the

²⁴ χ^ℓ is not a separate object from the LTF response; $\rho(\cdot)$ bundles the decision together with the terms, so evaluating it at the equilibrium FTF date — $\rho(t_{s1}^*, \omega)$ — gives $\chi^\ell(\omega)$, $t_{s2}^*(\omega)$ and $C_{s2}^*(\omega)$.

²⁵ That is, FTF cash and S1 penalty, plus (when the LTF settles) LTF cash and the S2 penalty, or (when the LTF litigates) the brand's LTF litigation costs.

filer's ex-ante value at the entry stage is

$$V_e^{piv}(K, N) = \beta^{t_e} \left[\frac{1}{K} \sum_{M=1}^K B_{K-1, 1/K}(M-1) V^f(\omega_M) + \frac{K-1}{K} \sum_{M=0}^{K-1} B_{K-1, 1/K}(M) V^\ell(\omega_M) \right]$$

When $K = 0$, there is no challenge and no settlement stage: the patent runs to T and all N generics enter at expiration. Writing τ^{exp} for this no-challenge timeline – structure $\omega = (0, 0, N)$, launch dates $(t_f, t_\ell, t_p) = (T, T, T)$, and no AG ($a_\tau = 0$) – gives $V^B((0, 0, N)) = \Pi^B(\tau^{exp}, (0, 0, N))$.

Free entry: Each candidate entrant chooses only between filing a PIV and staying out. The postpatent count P is set exogenously, so $N = K + P$.

In the first stage, a large number of generics decide whether they want to enter by filing a Paragraph IV certification. I assume all generic firms are ex-ante identical.

Generics keep applying for entry until the value from entering no longer exceeds the cost of entry. More precisely, the equilibrium number of PIV filers K^* is determined by the following inequality:

$$V_e^{piv}(K^*, N^*) \geq \kappa_{piv} > V_e^{piv}(K^* + 1, N^* + 1)$$

where κ_{piv} is the full sunk cost of filing a Paragraph IV certification and filing an ANDA. Intuitively, the value function of entry is declining in the number of entrants – more entrants mean more competition in the market and lower per-period profits. At the equilibrium K^* , current entrants can cover their entry cost, but an additional entrant would stop that from happening. This follows the free-entry condition in [Alam and Conti \(2026\)](#) and [Igami \(2018\)](#).

4 Discussion

Symmetric bargaining outcomes within a settlement stage: Fundamentally, this assumption is made to reduce the dimensionality of the bargaining problem while preserving the key economic mechanisms. Allowing each negotiation to resolve differently would introduce partial settlements and require additional assumptions about the order of negotiations, the brand's ability to discriminate among otherwise identical firms, strategic holdout, the effect of each agreement on the remaining firms' threat points, etc. It is unclear how these mechanisms can be modeled in my setting.

While the modeling assumption here is more for simplification rather than something to be taken at face value, there are multiple underlying negotiation models that can generate such symmetric bargaining outcomes. First, the outcomes from this symmetric bargaining game are identical

to a specific form of collective bargaining between the brand and all generics in a stage.²⁶ Second, we can think of the brand as bargaining with a representative generic, and all other generics accept the terms being negotiated or the subsequent litigation choice. Third, since all generics within a settlement stage are assumed to be symmetric, we can think that it is natural they all agree to the same terms, with additional assumptions ruling out strategic behavior and belief formation.

In short, it is difficult to have a reasonable model of within-stage bargaining asymmetry without introducing tremendous complication. Instead, this simplifying assumption allows me to retain the key economic tradeoffs in my setting. I therefore abstract from within-stage bargaining heterogeneity and focus on the distinction between first filers and later filers, and the sequential interaction between the two settlement stages.

Comprehensive litigation following Stage-1 breakdown: I assume that if the brand and the FTFs cannot reach a mutually acceptable settlement, then all PIV challengers litigate. Through this assumption, I effectively treat the FTF negotiation as the brand’s decision on whether to resolve all patent disputes through settlement or litigation. This makes sense, as FTFs challenge first and possess the greatest commercial leverage because they may benefit from regulatory exclusivity. A failed FTF negotiation may therefore be interpreted as a molecule-level litigation posture by the brand. The model thus assumes that the brand maintains this posture toward the remaining challengers rather than reopening settlement negotiations with the LTFs. In that sense, I use the comprehensive litigation branch as a reduced-form representation of the subsequent legal activity.

Furthermore, this aligns well with reality, as FTF litigation followed by LTF settlement is not something I have encountered in the literature; usually, FTFs gain large settlement offers and LTFs settle for similar/worse terms. If it does happen it is likely to be rare.

This assumption also allows me to avoid making unclear modeling choices. This is because i) a FTF litigation victory may lead to very early entry, so it is unclear what the LTF settlement will be bargaining over; ii) the LTF settlement may be negotiated after the FTF litigation has concluded, so again it is unclear what the LTF settlement will be bargaining over, and this will further depend

²⁶See Appendix D for more details, but I present a simplified example below. Take the example of S2 (the same argument applies in S1). One formulation of collective bargaining between a brand and multiple LTFs could be written as:

$$[\text{GFT}_{s2}^B(t_{s2}, C)]^\alpha [L\text{GFT}_{s2}^\ell(t_{s2}, C)]^{1-\alpha}$$

Here the GFT_{s2}^ℓ of each individual LTF is multiplied by the number of LTFs L , and their overall bargaining parameter is $1 - \alpha$. Under this formulation, for every candidate pair (t_{s2}, C) , we can rewrite the Nash product as:

$$[\text{GFT}_{s2}^B(t_{s2}, C)]^\alpha [L\text{GFT}_{s2}^\ell(t_{s2}, C)]^{1-\alpha} = L^{1-\alpha} [\text{GFT}_{s2}^B(t_{s2}, C)]^\alpha [\text{GFT}_{s2}^\ell(t_{s2}, C)]^{1-\alpha}$$

Conditional on ω , the number of LTFs L is fixed. Hence, $L^{1-\alpha}$ is a strictly positive constant that is common to every candidate cash transfer and settlement date. Multiplying the Nash product by this constant changes its level but cannot change its maximizer. This formulation therefore produces the same (t_{s2}^*, C_{s2}^*) as my symmetric bargaining setup.

on whether the FTF has won or lost.

Predicting litigation: At the currently calibrated parameters, there is always some settlement term that results in positive gains-from-trade for all sides, so the model never predicts litigation as an equilibrium outcome. This is driven by my assumption of complete-information Nash-bargaining. The complete-information assumption implies that there will never be a bargaining breakdown – if there are no positive gains from trade then the agents will never turn up to discuss settlement (in my case, this will mean going to litigation), and if there are positive gains from trade then the agents will find some settlement terms to agree upon.

However, in reality, a fraction of patent challenges end up in litigation. One factor may be that the parameter values vary widely across drugs and cases, and litigated patent cases have more extreme parameter values than the average case. Since I calibrate the model, I have to stick to values that match observed averages in the prior literature, but it may be that cases going to litigation have substantially different parameter values.

The legal and theory literature rationalize negotiation breakdowns/litigation via asymmetric information or optimism. See Appendix A for a discussion on this.

If our aim is to get the model to fit the observed litigation rates, there may be alternative ways of extending the model to account for litigation without asymmetric information or optimism.²⁷

Post-patent generics: In this paper I set the number of generics choosing to enter without filing a PIV (i.e, those who will launch whenever the patent expires either naturally or through invalidation) exogenously.

The post-patent generics file their ANDAs a few years before the patent expiration, and any PIV certification is likely filed much earlier, so it is unclear how to model them together. Conceptually, we may think of post-patent generics as making their entry decisions closer to the actual patent expiration date and significantly later than after all PIV settlements have been negotiated. See [Alam and Conti \(2026\)](#) for a model of generic entry decisions without patent challenges.

Cash versus delay in value transfer: In my model, the brand and generic can transfer value through two channels: cash and delay. Cash is a direct transfer of value from the brand to the

²⁷For instance, after PIV filings have been made, there could be a random variable which stochastically dictates if a case will outright proceed to litigation (independent of downstream bargaining outcomes). This random variable is thus encapsulating in a reduced-form way the realization of unobserved shocks that can derail settlement negotiations.

This would allow the model to generate litigation as an equilibrium outcome even under complete-information and rational bargaining. The probability of this variable choosing outright litigation exogenously could be calibrated to match the observed litigation rates in the data.

That said, this does make the strong assumption that factors causing such “outright litigation” somehow do not affect settlement outcomes. It also does not allow litigation probability to vary by policies that reduce gains-from-trade of settling.

generic, while reduced delay/earlier entry is an indirect transfer of value that allows the generic to earn profits for more periods. Similar value can be transferred through different mixes of cash and delay. This raises the question, how is the model pinning down cash versus delay in value transfer?

In a simple setting with just one generic and no antitrust scrutiny, the brand and generic choose delay to maximize the joint surplus, then choose cash to split the surplus. Prior literature (e.g. [Hemphill \(2006\)](#)) predicts that this will lead to maximum delay (i.e. entry at patent expiration) and large cash transfer. This is because delaying an additional period raises the brand's per-period profits by more than it reduces the generic's per-period profits, so the joint surplus is maximized by delaying entry as much as possible.

My setting is more complicated, as there are multiple generics bargaining sequentially, and the antitrust scrutiny creates friction in the value transfer mechanisms. Despite that, at the currently calibrated parameters and with antitrust scrutiny zeroed out, the model predicts that the settlement launch date is close to the date of patent expiration, and the corresponding cash transfer is very large.²⁸

The antitrust scrutiny function also plays an important role in pinning down cash versus delay in value transfer. Naturally, the fact that it penalizes cash and delay pulls them back to levels observed in the real world. However, the relative differences in the penalties on cash versus delay (parameters ψ_1 and ψ_2 versus ψ_3 and ψ_4) pin down the relative mix of cash versus delay in value transfer. For instance, if the penalty on cash is much larger than the penalty on delay, then the model will predict that the brand and generic will transfer value more through delay than through cash.

See Appendix [A](#) for additional complexities in real-world patent challenges that I do not directly model, such as at-risk launch and licensing agreements.

Interpretation and usefulness of antitrust scrutiny function: Below I discuss the antitrust scrutiny function and how to interpret it.

First, I discuss what the antitrust scrutiny function is meant to capture. It reflects the expected antitrust-related costs associated with settlements involving large cash and delay. For a given settlement, with some probability it may lead to antitrust investigations, potential antitrust lawsuits, and potential legal penalties. The likelihood of these events and the magnitude of resulting costs are

²⁸While at the current parameter values, the delay is extended till patent expiration, that is not always the case. Surprisingly, when p_ℓ is low, then in market configurations with one FTF and all others LTFs, the settlement is that the FTF enters very early while all LTFs enter very late. The reason is that this allows the brand to pay LTFs less in cash. The LTF's disagreement payoff is stronger if the FTF delay is longer, because then it can invalidate the patent and have many FTF-free periods. But if the FTF enters early to begin with, then invalidating the patent would leave it with far fewer FTF-free periods. The presence of accelerator clause adds further wrinkles to this situation.

At the calibrated antitrust scrutiny parameters, I check each market configuration and do not find evidence of this happening.

increasing in the size of the cash transfer and the length of the delay. Thus, Ψ_{s1} and Ψ_{s2} capture the idea that as parties strike deals with more cash and delay, the expected cost from antitrust authorities rises.²⁹

Within the model, there is an advantage to having this function. As explained previously, it reins in cash and delay to amounts seen in publicly-reported settlement terms. It also helps adjust the mix of cash and delay for value transfer between brand and PIV filers.

I now discuss some miscellaneous issues about the antitrust scrutiny function.

First, the antitrust scrutiny reduces both parties' agreement payoffs but does not affect disagreement payoffs, making settlement less valuable. Thus, as Ψ_{s1} and Ψ_{s2} become larger, the advantage of settlement over litigation shrinks, eventually leading to litigation.

Second, the antitrust scrutiny scales with the number of settling firms, so settlements involving more firms face a larger penalty.³⁰ Each party to a stage's settlement bears this penalty in full, reflecting that an antitrust lawsuit must be defended by every firm involved.

Third, the no-AG commitment is not penalized by the antitrust scrutiny function. Given that it is withholding a product rather than transferring a measurable cash/delay, it is unclear how to penalize AG release. No-AG commitments are increasingly being scrutinized by antitrust authorities as being anticompetitive, so this is a caveat to keep in mind for the no-AG counterfactual.

Discussion of antitrust scrutiny elements: I now discuss the elements of the antitrust scrutiny function as well as the modeling decisions behind them.

First, I discuss the gap scrutiny that S2 incurs. The practical reason why the gap penalty is needed is without it, in my simulations the LTFs often negotiate a settlement date later than 2 quarters after the FTF date (usually between 3.5-5 quarters). In contrast, most publicly reported reverse settlements show LTFs entering right after the expiration of the 2-quarter Hatch-Waxman exclusivity.

There are multiple ways of interpreting this gap scrutiny, and not all have to do with antitrust. It may be that negotiated dates of FTFs and LTFs being significantly far apart may draw suspicion from antitrust authorities. It may also capture behavioral bargaining factors; for instance, once FTFs settle on a settlement launch date, LTFs may use that date to anchor their negotiations and thus get similar launch dates as well. While this is not Nash-bargaining, prior research has shown that behavioral factors often impact bargaining outcomes (e.g. [Keniston et al. \(2024\)](#)).

Next, I discuss the delay scrutiny incurred in both S1 and S2. Recall that the delay terms are

²⁹Empirically, the antitrust scrutiny function may capture more than just antitrust issues. The delay scrutiny may capture short-termism or risk aversion by negotiators, the cash scrutiny may capture financing constraints, etc. I do not try to disentangle these factors in the paper.

³⁰This is meant to reflect the reality that if a settlement holds off multiple generics from launching, the potential antitrust risk from authorities is higher.

anchored to expected launch date under litigation, where the probability weighting comes from the patent strength. Thus, drugs with stronger patents face less scrutiny even for large delays. The idea behind this is that antitrust authorities can look at patent challenge and get a sense of how valid/likely it is, versus whether it is just a “fishing expedition” by generic manufacturers (Hemphill and Lemley, 2011). If the underlying case for the PIV challenge is flimsy, then the antitrust authorities are unlikely to litigate even if the delay is very large. In contrast, if the challenge appears to be carefully worked out, then even a medium delay may result in high scrutiny.

In my simulations, I find that anchoring the delay term to expected launch date under litigation is not without consequence. In particular, it turns out to strongly influence how likely litigation is under high levels of cash scrutiny. Even if scrutiny of cash transfers is extremely high so that cash transfer is close to zero, firms still settle close to the expected launch date under litigation, which then sets the delay scrutiny close to zero while allowing the firms to avoid litigation costs and remove the uncertainty from litigation. However, if the anchor is set to something like 4 or 8 quarters, litigation becomes far more likely. Settling at expected launch date under litigation results in a high antitrust penalty from delay, and without cash transfers to offset it, PIV filers may prefer litigation over settlement.

Finally, I discuss the cash scrutiny incurred in both S1 and S2. Note that unlike the delay and gap terms, the cash term is not anchored to some benchmark value. Thus, the antitrust scrutiny kicks in as soon as any positive cash transfer is agreed upon. This has to do with the obfuscation issue described above – the cash amount may be obfuscated using additional settlement terms, so it is unclear what to anchor cash against. Instead, with the current setup, the cash term is penalized in a quadratic manner; this ensures that the authorities are more likely to let go of small cash transfers, but will scrutinize larger cash transfers more heavily as those are harder to fully disguise.

See Appendix A for further discussions on the following topics: alternative explanations for litigation (asymmetric information and optimism), additional complexities in real-world patent challenges not in my model (e.g. at-risk launches, licensing), sunk cost in sequential bargaining, etc.

5 Calibration

An issue with studying pharmaceutical settlements is lack of comprehensive data on settlements and terms. Patent settlements are generally confidential, and while they have to be reported to the FTC and DOJ, the agencies can in turn only report back aggregate measures in MMA reports

(such as total settlements, broad characteristics of all settlements in a year, etc).³¹ For a handful of settlements, I can find their terms from antitrust cases filed in court and reported in the news. Nevertheless this is an important policy setting to study. As a result, I try to reasonably calibrate the model parameters using prior literature and some publicly-reported settlement terms.³²

5.1 Cost parameters

Litigation costs: Public estimates of litigation costs by PIV filers comes from a report prepared for the Office of the Assistant Secretary for Planning and Evaluation ([Eastern Research Group \(2021\)](#)). It lists a table of cost estimates for drug patent litigation. In summary, the report states that the lead litigant faces costs around \$3.5 million, and any follow-on litigant (piggybacking off the lead litigant’s case) faces costs around \$1 million. Similar figures are reported in [American Intellectual Property Law Association \(2021\)](#), which reports survey results on median litigation costs of Hatch-Waxman patent infringement cases.

Note that these are accounting costs, not economic costs. As a result, they do not include the opportunity cost of management time spent on litigation, constraints, etc. So in my calibration, I think of \$3.5m as a lower bound for the litigant’s cost.

As a result, I set all litigation costs of all agents (brand, FTF, LTF) at \$4m. Recall that the LTFs face an approval bottleneck – this makes it very costly and uncertain for them to pursue patent litigation if FTFs settle. To account for this, I set the LTF litigation cost to be the same as FTFs’.

Entry cost κ_{piv} : Prior literature and public estimates ([Starc and Wollmann \(2023\)](#), [Gottlieb \(2016\)](#)) place the cost of filing an ANDA without a Paragraph IV certification between \$5 million and \$15 million. The cost of pursuing a Paragraph IV certification is likely higher than the lower-end estimates for an ANDA-only filing because it also includes the cost of preparing the certification, incurred litigation cost prior to settlement, and, in some cases, challenges to multiple patents. In addition, only a limited set of firms has the capability to pursue Paragraph IV challenges ([Hemphill, 2006](#)), implying further fixed or opportunity costs.

I would also like my choice of κ_{piv} to match model-predicted entry with the observed number of Paragraph IV filers per molecule. As reported in Section 2.4, the average number of filers was approximately 8.6 per molecule in 2018. I set κ_{piv} so that the model produces a comparable number

³¹Links to reports: <https://www.ftc.gov/enforcement/competition-matters/2020/10/mma-reports-no-tricks-or-treats-just-facts>, and <https://www.ftc.gov/enforcement/competition-matters/2025/01/reverse-payments-cash-quantity-restrictions-other-possibilities>

³²Another defense of such calibration is: an aim of this paper is to construct a model that may be used to simulate and predict the effects of settlements and potential policy changes. Therefore, it may be used to aid antitrust cases on reverse settlements by using different parameter values to adjust to the setting. The current calibration thus demonstrates how the model works and helps set baseline expectations for the model’s predictions.

of filers. This yields an all-in expected entry cost of $\kappa_{piv} = \$12$ million.

5.2 Timing parameters

For calibrating the timing parameters, I use the following facts.

1. Timing of the first PIV filing: [Kannappan et al. \(2021\)](#) find a median of 5.2 years from brand approval to the PIV filing. [Hemphill and Sampat \(2011\)](#) find an increase in the number of challenges filed within the first 5 years of drug approval. Note that there is a 4-year lower bound in most cases: for a drug subject to five-year “New-Chemical-Entity exclusivity”, a Paragraph IV certification may generally be submitted no earlier than four years from the date of NDA approval.
2. Duration of patent litigation: [Eastern Research Group \(2021\)](#) combine multiple sources and state that the time from PIV filing to court decision ranges from 30-46 months (10-15 quarters).
3. Patent life remaining at PIV filing: [Hemphill and Sampat \(2011\)](#) find that brand drug products have approximately 12-14 years of post-approval patent life.
4. Gap between first-filer and later-filer entry: I could find only indirect evidence on this fact. [Hemphill and Lemley \(2011\)](#) study all 180-day exclusivity awards between 2005-2009, and find bunching of later-filer approvals at the end of the 180-day exclusivity period. Specifically, they note: “...when the exclusivity associated with the settling generic firm expired, there were immediate approvals of at least one other generic firm.” Furthermore, [Hemphill \(2006\)](#) provides descriptive evidence of a number of pay-for-delay settlements, many of which see later filers entering soon after the Hatch-Waxman exclusivity period.

I use these estimates to calibrate the timing parameters of the model:

1. t_e : This measures the time from a generic’s decision to pursue a Paragraph IV certification to the start of settlement negotiations. Regarding the former, this differs from the filing date because filing requires the generic to reverse engineer the drug, prepare an ANDA, find issues with the original patent, and complete the Paragraph IV certification. I assume that these steps take six quarters. After filing, the brand may invoke a 30-month stay, which delays negotiations by an additional 10 quarters. I therefore set $t_e = 16$ quarters.
2. T_f^{inv} and T_ℓ^{inv} : The time from a Paragraph IV filing to a court decision is typically 10–15 quarters. A 30-month stay accounts for 10 quarters of this period. It is reasonable to assume

that settlement negotiations begin during/after the stay and leave two quarters before resolution; accordingly, I set $T_f^{inv} = 2$ quarters. For the later-to-file generic, additional delay arises from later PIV submission and the approval bottleneck. I therefore set $T_\ell^{inv} = 6$ quarters.

3. T : The variable T measures the gap between settlement negotiations ending and the original patent expiration date. From above, we know that there are about 50 quarters of patent life when the brand launches. Subtracting 16 quarters from that to the first PIV filing (due to the NCE exclusivity) plus the 10-quarters it takes between filing and settlement negotiation wrapping up leaves me with 24 quarters between settlement negotiation and patent expiration. However, an increasingly common phenomenon for drugs with high sales is that brand drug manufacturers can find ways to extend patent life by filing additional patents, getting approval for new and rare indications, etc. Thus, I err on the side of being expansive and assume that in the absence of PIV filings, the brand may have been able to keep out generics for an additional 3 years (12 quarters), leaving us with $T = 24 + 12 = 36$ quarters.

Following [Alam and Conti \(2026\)](#), I assume that the generic market converges to zero profits 8 years after original patent expiration date, so I set $\bar{T} = 68$ quarters. Since the Hatch-Waxman exclusivity period is two quarters, I set $\Delta_{hwe} = 2$.

5.3 Other parameters

Per-period profits and prices: I use reported results from [Alam and Conti \(2026\)](#) to calculate per-period profits and prices. [Alam and Conti \(2026\)](#) contains descriptive regressions of market share, prices, and revenues on number of competitors, as well as an estimated range for marginal costs, using data from IQVIA National Sales Perspective. I combine these to predict per-period profits for the brand and generic firms, as well as their prices. I reproduce these regressions from [Alam and Conti \(2026\)](#) in Appendix E.

Patent strength (p_f and p_ℓ): These parameters determine how likely a patent is to be upheld if fully litigated. Perhaps unsurprisingly, it is difficult to find public estimates of general patent strength in pharmaceuticals, so I have to rely on indirect evidence. [Jacobo-Rubio et al. \(2020\)](#) find that of all pharmaceutical patent cases going to court, about 60% of the cases are won by the brand manufacturer. Therefore, I calibrate the baseline patent strength to 65%. Since LTFs face the approval bottleneck, which is both very costly and highly uncertain, I set the LTF to have a lower baseline probability of litigation success (i.e. higher probability of patent upheld) with $p_\ell = 0.75$.

Since this parameter is calibrated with indirect evidence, I conduct robustness checks by running counterfactuals for patent strengths from 0.5 to 0.8, and obtain similar qualitative results.

Bargaining parameter: There is no public estimate of bargaining parameters for negotiations between brand manufacturer and PIV filers, so I need to use indirect estimates instead. [Dubois et al. \(2022\)](#) estimate Nash-bargaining between drug manufacturers and Canada. For on-patent branded drugs, they find a range of bargaining parameters with a median of 0.71. For off-patent branded drugs and generics, the bargaining parameters are much lower. [Tunçel \(2025\)](#) estimates Nash-bargaining between drug manufacturers and French regulator, and finds bargaining parameters between 0.79-0.93.

I set the brand's bargaining parameter to be 0.7. Since this parameter is calibrated with indirect evidence and bounds, I conduct robustness checks by running counterfactuals for bargaining parameters from 0.55 to 0.75, and obtain similar qualitative results.

Antitrust scrutiny parameters: I choose these parameters so that the model-predicted cash transfers and delays are roughly in line with the publicly-reported settlement terms in Appendix B. Since reported settlements are likely to be biased towards larger and more suspicious settlements with larger cash transfers, I do not want to match the model-predicted values to the observed pay-for-delay settlements. Instead, I choose parameters so that my model-predicted values partly undershoot the cash transfers. I do try to have my model-predicted values be close to the observed delays.

In the end, I set the antitrust scrutiny parameters to $\psi_1 = 0.0$, $\psi_2 = 0.002$, $\psi_3 = 0.0$, $\psi_4 = 0.2$, $\psi_5 = 0.0$, and $\psi_6 = 0.1$ (with the cash transfers expressed in millions). Note that this means I have non-zero parameters only on the quadratic terms of the scrutiny function, and zero parameters on the linear terms. This is because I want to penalize larger cash transfers and delays more heavily than smaller ones, which seems to be in line with how antitrust authorities operate. It also lets me avoid having to take a stand on the parameters for the linear terms.

This raises the question of why I do not use a moment-matching estimator (like the Simulated Method of Moments) to estimate the antitrust scrutiny parameters. The reason is that there is heterogeneity in the other parameters across drugs, and without accounting for that heterogeneity, it is not clear how informative such an exercise would be. In particular, i) patent-strength is likely to vary tremendously across drugs; ii) I do not have estimates of per-period profits for each drug, and the per-period profits are likely to vary tremendously across drugs as well, iii) different drugs are litigated at different times relative to the start of their patent life. Since I pin these parameters at some baseline numbers and ignore heterogeneity, there should be serious misspecification issues if I try to estimate the antitrust scrutiny parameters using moment-matching.

I conduct robustness checks by varying ψ_2 (cash), ψ_4 (delay), and ψ_6 across a range of values around the baseline, then compute counterfactuals, and obtain similar qualitative results.

Remaining parameters: I set the number of postpatent generics $P = 2$. I also experiment with $P = 3$ and 4, and obtain similar results. [Alam and Conti \(2026\)](#) finds the vast majority of molecule-formulations to have between 1-11 generics, with the 90th percentile being 11. Since my model is calibrated to produce around 7-9 PIV filers, $P = 2-4$ gets us near the upper limit of this range. This is reasonable as I choose to focus on larger markets with more generics, as explained in Section 5.4. I set discount factor $\beta = 0.98$ following prior literature on dynamic games.

5.4 Choice of markets

Since I calibrate per-period profits using [Alam and Conti \(2026\)](#), and [Alam and Conti \(2026\)](#) defines markets as the molecule-formulation combination, I also define markets as the molecule-formulation combination.³³ Recall that I use their reported revenue regressions to predict per-period profits, so the variation in markets in this paper mainly comes from the variation in values of fixed effects in that regression. A larger value of fixed effect implies a larger per-period profit, and thus a larger “market size”. To better pin our intuition, we can think of the revenue regression fixed effect or market size as a proxy for the brand’s monopoly profits.

My aim is to select a subset of markets and calibrate the model’s parameters to match observed pay-for-delay settlement values and the number of Paragraph IV filers. For links to settlement data and summary statistics, see Appendix B.

Thus, if I rank markets (fixed effect values) by brand’s monopoly profits, the question is what percentile of market sizes I should use as the baseline to calibrate my model.

Since the prior literature finds that molecules with larger market sizes are more likely to face Paragraph IV challenges, I do not want to match the median market’s values to the observed pay-for-delay settlements. There is also a long left tail of market sizes. Instead, I would like to skew towards larger markets. Thus, I pick markets around the 80th percentile range.³⁴ I also run my counterfactuals for markets somewhat larger and smaller than the 80th percentile and obtain the same qualitative results.

I also focus only on markets where the brand will find it profitable to release an AG. That is, for each possible number of generics present in the market, adding an AG raises the brand’s profits. Markets not satisfying this condition are filtered out.³⁵

³³ [Alam and Conti \(2026\)](#) select “molecule-formulations that saw its first generic entrant after 2004 and before 2016, which are not biologics, for which we can identify presence or absence of AG with certainty, and those that are not surrounded by strong external circumstances (e.g. media/public outrage, serious or repeated lawsuits)”. This leaves them with 241 molecule-formulations, of which 58 are injectables. The regressions that I use for calibration are run on non-injectable molecule-formulations.

³⁴ That is, 80th percentile from the fixed effect values of revenue regression in [Alam and Conti \(2026\)](#).

³⁵ Why do I perform this check? A no-AG commitment transfers value only if launching an authorized generic benefits the brand and harms the settling generic. In that case, the brand gives up something valuable by promising not to launch the AG. The no-AG commitment can then substitute for a cash payment. If launching an AG would not

6 Counterfactuals

With the model set up and parameters calibrated, I study the effects of various counterfactuals on equilibrium outcomes.

Throughout, I use the term “entry incentive” to mean V_e^{piv} . This is because, holding the number of PIV filers fixed, an increase in V_e^{piv} makes it more likely that a new PIV filer will enter. Thus, an increase in V_e^{piv} signifies an increase in the entry incentive, and weakly greater PIV filings.³⁶ The phrases “FTF/LTF/brand payoffs” refer to $V^f(\omega)$, $V^\ell(\omega)$, and $V^B(\omega)$ respectively.³⁷ I denote “total reverse payment” to mean the total amount paid to FTFs and LTFs combined.

I will note that V_e^{piv} is a probability-weighted function of FTF and LTF payoffs, where the probability weights are determined by the number of PIV filers (aiming to roughly mimic a first-to-file race). In some counterfactuals, FTF and LTF payoffs may move in opposite directions. In such cases, my assumption of probability-weights influences the direction of V_e^{piv} , and thus the entry incentive. A different assumption on probability weights may lead to different conclusions about the entry incentive.

I gauge the impact of policies on consumer welfare by studying how they affect “earliest generic entry dates” and “total generics”. An earlier generic entry date means that prices drop from monopoly levels earlier. More generics means greater competition, pushing down prices.³⁸ Together these should lead to improvements in consumer welfare. Finally, I focus on non-brand prices rather than brand prices to gauge consumer welfare. From prior literature (such as [Alam and Conti \(2026\)](#)) we know that once generic entry forces down prices, the majority of consumers switch over to non-brand products.

Exclusivity awarded only for winning litigation: Hatch-Waxman exclusivity is given to the first filing generic, rather than the first generic to market the product or to invalidate the patent. This has raised some criticism in policy circles. The argument goes that if a first-filer strikes an anticompetitive settlement with a brand, it is actually rewarded with the Hatch-Waxman exclusivity period, which in turn serves as an approval bottleneck and holds up LTFs.

I study what happens if the Hatch-Waxman exclusivity is awarded to a FTF only if it litigates and invalidates the patent, but not if it settles. This counterfactual mimics the policy proposal of “Earned Exclusivity” put forth in [Hemphill and Lemley \(2011\)](#). I find that it leads to earlier

benefit the brand, withholding it costs the brand nothing. The no-AG promise is then not an economic concession.

³⁶Recall the free entry condition $V_e^{piv}(K^*, N^*) \geq \kappa_{piv} > V_e^{piv}(K^* + 1, N^* + 1)$. If a policy change leads to V_e^{piv} increasing for a given K , it raises the chances of a $K + 1$ th generic entering and still satisfying the free-entry condition.

³⁷Intuitively, for each agent, these refer to the the discounted profit streams plus any cash transfers paid out/received, minus antitrust scrutiny cost if settling or litigation cost if litigating.

³⁸The fact that increasing generics leads to lower and lower prices is verified in [Alam and Conti \(2026\)](#) (see Table 6) and also mentioned in prior literature.

settlement entry for FTFs and LTFs, and cash transfers generally decline.

In my simulations, the total PIV filings remain unchanged. The change in entry incentives is small and ambiguous; this is because FTF's payoff always declines, but sometimes LTFs' payoff goes up. The LTFs' gains come from the fact that removing Hatch-Waxman exclusivity for FTF settlement lets it negotiate an earlier launch date.

Consumer welfare improves if number of PIV filings stays the same – the first generic as well as the LTFs enter earlier, leading to an earlier decline from brand's monopoly prices.

No-AG commitment: As antitrust scrutiny of cash transfers has risen, brand manufacturers have found alternative ways of transferring value to PIV filers. One such way is for the brand to withhold its Authorized Generic if the PIV filers agree to settle. This removes an additional source of competition during the exclusivity period and also potentially in the long run. What is the impact of such a commitment on market outcomes and consumer welfare? To study this, I contrast the outcomes under inclusion of no-AG commitment in settlement terms with outcomes under unconditional AG release.

With a no-AG commitment included in the settlement terms, entry incentives are larger, leading to usually one additional PIV filing. The settlement launch dates remain quite similar with or without the no-AG commitment. Total reverse payment to generics declines, demonstrating how the no-AG commitment effectively substitutes for cash transfer in such reverse settlements (in the absence of antitrust scrutiny on such commitments).

Why are entry incentives larger? The brand reduces competition during the Hatch-Waxman exclusivity period, which is attractive for FTFs. There is also now no competition from AGs in later periods, which is beneficial for LTFs as well. On the other hand, the brand pays less cash to the FTFs, and total reverse payment declines. On net, the expected payoff for PIV filers rises. Thus entry incentives are larger, leading to more PIV certifications.

The no-AG commitment generally has a negative impact on consumer welfare. The first generic enters at roughly the same time, so the market is not opening up earlier. If the number of PIV filings stays the same, then price rises as generics no longer face additional competition with the AG, especially during the Hatch-Waxman exclusivity period. If the number of PIV filings rises by one to offset the AG entry, then long run prices are similar as in the case with unconditional AG release. Short run prices will be similar if the new PIV filer enters as a FTF. But if the new PIV filer enters as a LTF, the market misses additional AG competition during the Hatch-Waxman exclusivity period, which leads to higher prices during that period.

Accelerator clause: When accelerator clause is included in settlement terms, entry incentive declines by a small amount, and in my simulations it does not decline by enough to change the

number of PIV filers. Total reverse payment declines. The negotiated launch dates stay similar.

Entry incentive is lower because while the FTFs' expected payoff rises by a small amount, LTFs' expected payoff falls by more. The LTF's disagreement payoff is lower because invalidation no longer buys it some periods without the FTF; in turn, this means the negotiated C_{s2} in Nash-bargaining is lower. The FTF's payoff rises somewhat because the lower C_{s2} leads to a greater surplus being split in $S1$, and that leads to the negotiated C_{s1} being generally slightly higher.

If there is no change in entry, then consumer welfare remains unchanged, as there is no change in the launch dates or number of competitors.

Raising antitrust scrutiny of cash transfers: Reverse settlements have been a source of concern for antitrust authorities for a long time. As more supportive court decisions occurred, antitrust scrutiny of reverse settlements involving large payments ramped up, so what is the equilibrium effect?

As antitrust scrutiny of cash transfer rises, FTFs and LTFs negotiate earlier settlement launch dates. Entry incentive declines; for an increase in cash scrutiny of 2-10 times, total PIV filings usually declines by one. Total reverse payment, C_{s1} , and C_{s2} decline. Payoffs of FTFs and LTFs generally decline.

From consumers' perspective, the first generic enters earlier, which leads to an earlier decline from brand monopoly pricing. However, the number of generic competitors may be weakly fewer, which may lead to a smaller decline in price in the long term. Thus, the net effect on consumer welfare is ambiguous.

This counterfactual lines up with [Edlin et al. \(2013\)](#). There, the authors posit that larger reverse settlements (in their case, payment to generics above avoided litigation cost) lead to greater delay, thus diminishing the expected periods of competition.³⁹ In my counterfactual, I also find that as cash scrutiny rises, cash transfer declines, and PIV filers negotiate earlier launch dates.

Cash-transfer ban and R&D incentives: If cash transfers are banned, does the relative return from a strong patent increase? This may tilt R&D incentives towards drugs with stronger patents.⁴⁰

While we cannot study R&D directly as it is not part of our model, we can see how lifetime payoffs for different products change due to policy. The intuition is that if the lifetime payoffs for a particular type of product rises relative to others, then the brand manufacturer has more incentive to

³⁹More precisely, the authors compare the settlement launch date with the "expected launch date upon litigation", where the latter is a function of patent strength. They use simple algebra to show that if the brand's payment to generics exceeds avoided litigation costs, then the settlement launch date moves further beyond the expected launch date upon litigation. Furthermore, the larger is the net cash transfer, the longer is the likely delay.

⁴⁰Note that even with a cash ban, in my current setup nearly all PIV filers settle at a date somewhat close to the expected launch date under litigation. If the delay scrutiny's benchmark date is set to a lower number, the likelihood of litigation increases.

spend on R&D to develop that product. More specifically, if lifetime payoffs of strong-patent drug rises relative to weak-patent drug as a result of policy, then brand manufacturer has an incentive to shift more R&D resources towards discovering strong-patent drugs.

Let “strength premium” be the difference in the brand’s payoff from having a strong patent ($p_f = p_\ell = 0.7$) compared to a weak patent ($p_f = p_\ell = 0.4$). The “change in the strength premium” captures how a policy affects this difference relative to the baseline. An increase makes strong patents relatively more attractive and potentially shifts R&D incentives toward products likely to receive strong patent protection. A decrease shifts incentives in the opposite direction.

I find the ban on cash transfers to have an ambiguous effect on the change in strength premium. That is, in some markets, it tilts R&D incentives towards stronger patents; in others, it tilts them towards weaker patents.

What drives this result? To see why, first note that weaker patents see more PIV certifications and higher reverse payments than strong patents, which reduce the brand’s payoff. Next, I explain how banning cash transfer has multiple effects:

1. It reduces reverse payments, leading to savings for the brand. Because these payments are initially larger for weak-patent drugs, weak-patent brands generally realize greater cash savings. This channel reduces the strength premium.
2. It changes settlement launch dates. Without cash transfers, the parties substitute toward earlier generic entry. Entry moves earlier for both weak- and strong-patent drugs, but neither type consistently gives up more delay. The relative effect depends on how much cash the brand saves for each quarter of exclusion it surrenders, which in turn depends on the molecule’s product-market profits.
3. It weakly reduces the equilibrium number of PIV filers for both patent types. This raises brand payoffs, but the relative benefit is not consistent across markets. Moreover, the entry response is discrete: the number of filers changes only when the marginal PIV filer’s expected payoff falls below κ_{piv} .

Because the cash-saving, settlement-timing, and entry effects need not favor the same patent type, a ban on cash transfers can either increase or decrease the strength premium.

Hatch-Waxman exclusivity length: Recall that the 180-day Hatch-Waxman exclusivity is meant to incentivize generics to file PIV certifications against a patent. There have been some concerns about the length of this exclusivity period regarding whether it is too short or too long.

I study what happens if Hatch-Waxman exclusivity length rises from 1 to 4 quarters. As HWE length rises, FTF negotiates earlier settlement entry, and LTF negotiates generally similar

or slightly later settlement entry. Total reverse payment usually (but not always) falls. At the current calibration, while entry incentive rises somewhat, it usually (but not always) does not rise by enough to create more PIV filings.⁴¹

The effect on consumer welfare is likely positive. The first generic enters earlier, plus if there is an increase in PIV filings then the increased generic competition will lead to a greater price decline. These would benefit consumer welfare. While the LTFs may enter later, that delay is very small, and so does not significantly slow down the price decline.

Varying patent strength: I study how equilibrium outcomes vary with patent strength of the underlying drug. As previously discussed under the concept of probabilistic patents, some patents are stronger than others, and therefore are more likely to be upheld in court. Such patents have greater “patent strength”. In my model this is governed by the parameters p_f and p_ℓ , and higher values of these parameters signify greater patent strength. I find the following :

1. As patent strength goes up,⁴² the incentive to file PIV certifications declines, leading to fewer PIV filers. Both FTFs and LTFs choose later settlement dates. The brand’s payoff rises.

The intuition for this is as follows. A stronger patent is less likely to be invalidated in court, which improves the brand’s disagreement payoff while worsening that of generics. This allows the brand to negotiate later settlement dates. The delay-scrutiny’s benchmark date also shifts back, allowing the brand to delay PIV filers further without incurring greater antitrust scrutiny. The later entry reduces PIV filers’ lifetime profits, reducing entry incentives.

From consumers’ perspective, for stronger patents, the first generic thus enters later, and there are potentially fewer generic competitors.

2. For very low patent strengths (p_f and p_ℓ between 0.005 and 0.1), number of PIV filers is larger, negotiated settlement dates by both FTFs and LTFs are very early. There is no litigation.
3. For very high patent strengths (p_f and p_ℓ between 0.85 and 0.9), number of PIV filers is smaller, and there is no litigation. The negotiated settlement dates are very late and almost at patent expiration.

⁴¹FTFs’ payoff rises, but LTFs’ payoff falls somewhat due to the slightly later entry.

⁴²I set $p_f = p_\ell$ and vary them from 0.3 to 0.7.

7 Conclusion

This paper develops and calibrates a model of Paragraph IV challenges, patent settlements, and litigation. Potential generic manufacturers decide whether to challenge a patent. The brand then bargains in sequence with FTFs and LTFs over launch dates and cash transfers. If bargaining fails, the relevant firms litigate. The resulting launch dates determine product-market profits, which in turn shape the incentive to file a challenge. I use the model to conduct various policy counterfactuals, and show how such policies affect consumer welfare by influencing both settlement terms and PIV filing incentives.

References

- Alam, Rubaiyat and Rena M Conti (2026) “Entry Delays and Fighting Brands: Evidence from Generics and Authorized Generics.”
- American Intellectual Property Law Association (2021) “Report of the Economic Survey 2021: Relevant Excerpts.”
- Asker, John and Volker Nocke (2021) “Collusion, mergers, and related antitrust issues,” in *Handbook of industrial organization*, Vol. 5: Elsevier, 177–279.
- Bokhari, Farasat AS, Franco Mariuzzo, and Arnold Polanski (2020) “Entry limiting agreements: First-mover advantage, authorized generics, and pay-for-delay deals,” *Journal of Economics & Management Strategy*, 29, 516–542.
- Branstetter, Lee, Chirantan Chatterjee, and Matthew J. Higgins (2016) “Regulation and welfare: evidence from paragraph IV generic entry in the pharmaceutical industry,” *The RAND Journal of Economics*, 47, 857–890, DOI: <https://doi.org/10.1111/1756-2171.12157>.
- Bulow, Jeremy (2004) “The gaming of pharmaceutical patents,” *Innovation policy and the economy*, 4, 145–187.
- Ding, Yucheng and Xin Zhao (2019) “Pay-for-delay patent settlement, generic entry and welfare,” *International Journal of Industrial Organization*, 67, 102532, DOI: <https://doi.org/10.1016/j.ijindorg.2019.102532>.
- Drake, Keith M and Thomas G McGuire (2025) “Using stock price movements to estimate the harm from collusive drug patent litigation settlements,” *Journal of Health Economics*, 103054.

- Dubois, Pierre, Ashvin Gandhi, and Shoshana Vasserman (2022) “Bargaining and International Reference Pricing in the Pharmaceutical Industry,” *Working paper*, 2020, DOI: [10.3386/W30053](https://doi.org/10.3386/W30053).
- Eastern Research Group, Inc. (2021) “Cost of Generic Drug Development and Approval,” December.
- Edlin, Aaron, Scott Hemphill, Herbert Hovenkamp, and Carl Shapiro (2013) “Activating actavis,” *Antitrust*, 28, 16.
- Edlin, Aaron, Scott Hemphill, Herbert Hovenkamp, and Carl Shapiro (2015) “The actavis inference: Theory and practice,” *Rutgers UL Rev.*, 67, 585.
- Edlin, Aaron S, C Scott Hemphill, Herbert Hovenkamp, and Carl Shapiro (2014) “Actavis and error costs: A reply to critics,” *The Antitrust Source*, October.
- Farrell, Joseph and Carl Shapiro (2008) “How strong are weak patents?” *American Economic Review*, 98, 1347–1369.
- Federal Trade Commission (2010) “Pay-for-Delay: How Drug Company Pay-Offs Cost Consumers Billions,” staff study, Federal Trade Commission.
- Feldman, Robin and Evan Frondorf (2017) *Drug wars: How big pharma raises prices and keeps generics off the market*: Cambridge University Press.
- Ghili, Soheil and Matt Schmitt (2016) “A framework for estimating damages in reverse payment cases,” *Antitrust LJ*, 81, 873.
- Gottlieb, Scott (2016) “How Obama’s FDA Keeps Generic Drugs Off the Market,” *Missouri Medicine*, 113, 444.
- Grabowski, Henry G., Genia Long, Richard Mortimer, and Mehmet Bilginsoy (2021) “Continuing Trends in U.S. Brand-Name and Generic Drug Competition,” *Journal of Medical Economics*, 24, 908–917, DOI: [10.1080/13696998.2021.1952795](https://doi.org/10.1080/13696998.2021.1952795).
- Gratz, Linda (2012) “Economic analysis of pay-for-delay settlements and their legal ruling.”
- Hemphill, C Scott (2006) “Paying for delay: Pharmaceutical patent settlement as a regulatory design problem,” *NYUL Rev.*, 81, 1553.
- Hemphill, C. Scott and Mark A. Lemley (2011) “Earning Exclusivity: Generic Drug Incentives and the Hatch–Waxman Act,” *Antitrust Law Journal*, 77, 947–989.

- Hemphill, C Scott and Bhaven Sampat (2013) “Drug patents at the Supreme Court,” *Science*, 339, 1386–1387.
- Hemphill, C. Scott and Bhaven N. Sampat (2011) “When Do Generics Challenge Drug Patents?” *Journal of Empirical Legal Studies*, 8, 613–649, DOI: [10.1111/j.1740-1461.2011.01235.x](https://doi.org/10.1111/j.1740-1461.2011.01235.x).
- Hemphill, C Scott and Bhaven N Sampat (2025) “Patents, innovation, and competition in pharmaceuticals: the Hatch-Waxman Act after 40 years,” *Journal of Economic Perspectives*, 39, 27–52.
- Igami, Mitsuru (2018) “Industry Dynamics of Offshoring: The Case of Hard Disk Drives,” *American Economic Journal: Microeconomics*, 10, 67–101, DOI: [10.1257/MIC.20150256](https://doi.org/10.1257/MIC.20150256).
- Igami, Mitsuru and Kosuke Uetake (2020) “Mergers, innovation, and entry-exit dynamics: Consolidation of the hard disk drive industry, 1996–2016,” *The Review of Economic Studies*, 87, 2672–2702.
- Jacobo-Rubio, Ruben, John L. Turner, and Jonathan W. Williams (2020) “The Distribution of Surplus in the US Pharmaceutical Industry: Evidence from Paragraph IV Patent-Litigation Decisions,” *Journal of Law and Economics*, 63, 203–238, DOI: [10.1086/707407](https://doi.org/10.1086/707407).
- Kannappan, Sunand, Jonathan J. Darrow, Aaron S. Kesselheim, and Reed F. Beall (2021) “The Timing of 30-Month Stay Expirations and Generic Entry: A Cohort Study of First Generics, 2013–2020,” *Clinical and Translational Science*, 14, 1917–1923, DOI: [10.1111/cts.13046](https://doi.org/10.1111/cts.13046).
- Keniston, Daniel, Bradley J Larsen, Shengwu Li, JJ Prescott, Bernardo S Silveira, and Chuan Yu (2024) “Splitting the Difference in Incomplete-Information Bargaining: Theory and Widespread Evidence From the Field,” *International Economic Review*, 65, 1911–1939.
- Kong, Yunmi, Bernardo S Silveira, and Xun Tang (2025) “Risk and information in dispute resolution: An empirical study of arbitration,” *Journal of Political Economy*, 133, 2794–2835.
- Lee, Robin S and Kyna Fong (2013) “Markov perfect network formation: An applied framework for bilateral oligopoly and bargaining in buyer-seller networks.”
- Lemley, Mark A and Carl Shapiro (2005) “Probabilistic patents,” *Journal of Economic Perspectives*, 19, 75–98.
- Merlo, Antonio and Xun Tang (2019) “Bargaining with optimism: Identification and estimation of a model of medical malpractice litigation,” *International Economic Review*, 60, 1029–1061.
- Mungan, Murat C (2013) “Reverse payments, perverse incentives,” *Harv. JL & Tech.*, 27, 1.

- Palikot, Emil and Matias Pietola (2023) “Pay-for-delay with settlement externalities,” *The RAND Journal of Economics*, 54, 387–415.
- Patel, Ankur N (2009) “Delayed Access to Genetic Medicine: A Comment on the Hatch-Waxman Act the Approval Bottleneck,” *Fordham L. Rev.*, 78, 1075.
- Reiffen, David and Michael R. Ward (2005) “Generic drug industry dynamics,” *Review of Economics and Statistics*, 87, 37–49, DOI: [10.1162/0034653053327694](https://doi.org/10.1162/0034653053327694).
- Scott Morton, Fiona (2013) “Pay-for-Delay,” *Competition Policy International*, 9, 128–136.
- Scott Morton, Fiona M. (1999) “Entry Decisions in the Generic Pharmaceutical Industry,” *The RAND Journal of Economics*, 30, 421, DOI: [10.2307/2556056](https://doi.org/10.2307/2556056).
- Shapiro, Carl (2003) “Antitrust limits to patent settlements,” *RAND Journal of Economics*, 391–411.
- Sieg, Holger (2000) “Estimating a bargaining model with asymmetric information: Evidence from medical malpractice disputes,” *Journal of Political Economy*, 108, 1006–1021.
- Silveira, Bernardo S (2017) “Bargaining with asymmetric information: An empirical study of plea negotiations,” *Econometrica*, 85, 419–452.
- Starc, Amanda and Thomas Wollmann (2023) “Does Entry Remedy Collusion? Evidence from the Generic Prescription Drug Cartel,” *American Economic Review*, DOI: [10.2139/SSRN.4076263](https://doi.org/10.2139/SSRN.4076263).
- Tunçel, Tuba (2025) “Should we prevent off-label drug prescriptions? Empirical Evidence from France,” *Review of Economic Studies*, 92, 2027–2070.
- Van de Wiele, Victor L, Jonathan J Darrow, and Aaron S Kesselheim (2021) “No parking here: a review of generic drug 180-day exclusivity and recent reform proposals,” *Yale J. Health Pol’y L. & Ethics*, 20, 131.

A Additional discussions

Alternative explanations for litigation: The law and economics literature has given two explanations for why settlements fail and lead to litigation – asymmetric information and optimism.

First, there may be asymmetric information between the brand and the generics about patent strength. For instance, the generic may have private information about the likelihood of successfully recreating the drug and challenging the patent. However, it is unclear how likely asymmetric information is in this setting. Both sides employ specialized legal teams to assess patents, and furthermore the PIV filers need to submit a completed form showing how they think the patent might be invalid/uninfringed, which the brand can look at before starting litigation. Generic manufacturers also specialize in reverse-engineering drugs.

Second, litigation may be caused by “optimism” by both sides. As explained in [Merlo and Tang \(2019\)](#), optimism in this setting means that both sides believe that they have a better chance of winning than they actually do; in a technical sense, their beliefs of winning add up to greater than one. [Merlo and Tang \(2019\)](#) find evidence of optimism in medical malpractice lawsuits. This issue of “optimism” is beyond the scope of this paper, but is a possible answer and may be a subject of future research.

Additional complexities in real-world patent challenges: I do not model “at-risk launch” by generics. An at-risk launch by a generic is when a PIV filer launches its product before patent litigation is fully resolved. I avoid this for two reasons. First, at-risk launches are relatively uncommon. Second, if a generic launches at-risk and the patent is later upheld, then the generic is liable for damages to the brand manufacturer. If the brand can recover its profits, it is not clear why there would be some net effect worth modeling from this setting.

Sometimes, a settlement agreement involves the brand licensing the litigated product to a generic for production. In this case, the generic can sell the product, but has to pay back a fee to the brand or is limited in quantity sold. I think of this as an indirect way of transferring cash from the brand to the generic, although the implications for drug prices to consumers are unclear. I therefore avoid explicitly modeling licensing agreements for simplicity.

Sunk cost versus contingent cost: The brand negotiates sequentially, first with the FTFs, then with the LTFs. This sequence creates an important asymmetry between the two payments.

By the time the LTF negotiation begins downstream after S1, the FTF settlement has already been agreed upon. The brand must make the FTF payment whether the LTF settles or litigates. The FTF payment is therefore a sunk cost at S2: it lowers the brand’s payoff from both settling and litigating with the LTF.

However, upstream in S1 when negotiating with the FTFs, the brand anticipates the subsequent costs in S2 if S1 results in settlement. These include the LTF cash payment and the associated antitrust penalty. Importantly, the expected LTF bill is a “contingent cost” of settling with the FTF: it is incurred when the FTF settles, but avoided when the FTF litigates and ends the game.

As a result, the FTF bargains over a surplus that has already been reduced by the expected downstream cost in S2. A larger downstream cost leaves less surplus available to divide between the brand and the FTF. Under Nash bargaining, each additional dollar of downstream cost reduces the negotiated C_{s1} . The downstream payment C_{s2} therefore partially crowds out the upstream payment C_{s1} .

Thus, the FTF's settlement payment is a sunk cost when the S2 bargain is reached, but the LTF's settlement payment is a contingent cost when the S1 bargain is being negotiated.

B Statistics on reported reverse settlements

I use online sources to find reverse settlements whose terms are publicly reported. While I am able to find about 50 settlements, many do not have reported cash transfers, or their transfers are disguised through various mechanisms. For 14 settlements, a cash amount was disclosed in a document, and I report the distribution of cash and delay for these settlements below.

I use multiple LLMs to sort through various reports on these settlements to find the cash amount and delay amount. The delay is meant to measure the time between settlement being announced and the settled date of entry.

That said, the terms reported are often complicated and hard to work through, so the terms I extract via LLMs should be treated as approximations. Note that actual settlement terms often transfer cash as installments over several years. The numbers reported below try to aggregate them into a single lump-sum.

External link to settlement terms: [link](#)

Table 2: Reverse settlements: payment, by sample.

Sample	N	Mean	Median	Min	Max
Reverse settlements	14	70.88	61.50	12.00	220.00

Table 3: Reverse settlements: entry delay, by sample.

Sample	N	Mean	Median	Min	Max
Reverse settlements	14	5.90	6.28	1.37	9.42

Table 4: Distribution of cash transfers

Decile	Payment
0%	12.00
10%	20.30
20%	35.40
30%	45.00
40%	60.00
50%	61.50
60%	70.20
70%	73.95
80%	88.13
90%	114.45
100%	220.00

Table 5: Distribution of entry delays

Decile	Delay
0%	1.37
10%	1.97
20%	2.81
30%	4.09
40%	6.19
50%	6.28
60%	6.33
70%	8.49
80%	8.96
90%	8.96
100%	9.42

C Model solution

C.1 Overall solution structure

The model is solved by backward induction. For a given number of PIV filers K :

1. Starting with S2, at every candidate FTF settlement date $t_{s1} \in \mathcal{T}_1$, I solve for the Nash-product maximizing settlement offers (t_{s2}^*, C_{s2}^*) or determine that no settlement is feasible and the LTFs will litigate. This yields the LTF response correspondence $\rho(t_{s1}, \omega)$.
2. Next I solve for the S1 problem. This requires me to solve for the Nash-product maximizing settlement offers (t_{s1}, C_{s1}) or to determine that no settlement is feasible and the FTFs will litigate. In this problem, the LTF response correspondence $\rho(t_{s1}, \omega)$ is an input to the S1 problem, as the negotiation accounts for LTF response.
3. After solving the S1 and S2 problems, I can calculate the conditional payoff values for the brand, FTF, and LTF. The latter two can then be appropriately probability-weighted to recover the conditional value for filing a Paragraph IV.

I loop over $K = 0, 1, 2, \dots$ calculating the conditional values of PIV entry until the free-entry condition is satisfied.

C.2 Details on computation of Nash Bargaining

I give an overview of the solution method for Nash bargaining, then write out the details.

In the Nash-bargaining, I solve for the optimal pair of date (in quarters) and cash. I use a discrete grid for date, with each period representing a quarter. I use a continuous choice variable for cash. For a given date, I find the Nash-product maximizing cash using first-order conditions. I do this for each possible date. The combination of date with corresponding optimal cash amount that maximizes the Nash product is chosen as the solution.

S2 Nash bargaining: For each candidate date t_{s2} , I take the derivative in C of the logged Nash-product to get the cash first-order condition.

$$\Phi_{s2}(C | t_{s2}, t_{s1}, \omega) \equiv \frac{\alpha}{\text{GFT}_{s2}^B(C)} \frac{\partial \text{GFT}_{s2}^B}{\partial C} + \frac{1 - \alpha}{\text{GFT}_{s2}^\ell(C)} \frac{\partial \text{GFT}_{s2}^\ell}{\partial C} = 0,$$

where

$$\frac{\partial \text{GFT}_{s2}^B}{\partial C} = -L - \frac{\partial \Psi_{s2}}{\partial C}, \quad \frac{\partial \text{GFT}_{s2}^\ell}{\partial C} = 1 - \frac{\partial \Psi_{s2}}{\partial C}.$$

The candidate set of admissible cash values at date t_{s2} consists of (i) every interior root of $\Phi_{s2}(\cdot | t_{s2}, t_{s1}, \omega)$ on $[0, \bar{C}]$ that is a local maximum (second-order condition) and satisfies positive gains-from-trade for all, $\text{GFT}_{s2}^B > 0$ and $\text{GFT}_{s2}^\ell > 0$; and (ii) the zero-cash corner $C_{s2} = 0$, evaluated at every date whether or not an interior root exists there and admitted whenever it satisfies the positive gains-from-trade for all.

S1 Nash bargaining: For each candidate date t_{s1} , the LTF response $\rho(t_{s1}, \omega)$ is held at its stage-2 solution. I take the derivative in C of the logged Nash-product to get the cash first-order condition.

$$\Phi_{s1}(C | t_{s1}, \omega) \equiv \frac{\alpha}{\text{GFT}_{s1}^B(C)} \left(-F - \frac{\partial \Psi_{s1}}{\partial C} \right) + \frac{1-\alpha}{\text{GFT}_{s1}^f(C)} \left(1 - \frac{\partial \Psi_{s1}}{\partial C} \right) = 0,$$

where $\partial \Psi_{s1} / \partial C = F(\psi_1 + 2\psi_2 C)$.

The candidate set of admissible cash values at date t_{s1} consists of (i) every interior root of $\Phi_{s1}(\cdot | t_{s1}, \omega)$ on $[0, \bar{C}]$ that is a local maximum (second-order condition) and satisfies positive gains-from-trade, $\text{GFT}_{s1}^B > 0$ and $\text{GFT}_{s1}^f > 0$; and (ii) the zero-cash corner $C_{s1} = 0$ evaluated at every date whether or not an interior root exists there and admitted whenever it satisfies positive gains-from-trade for all.

Throughout, I set $\bar{C}$ to be an extremely large number to make sure that the cash amount is not artificially constrained by the upper bound.

D Collective interpretation of the generic-side Nash bargain

At each settlement stage, I write the Nash product using the gains-from-trade of one representative generic firm. This section explains why the resulting settlement terms are identical to those obtained when the generic cohort bargains collectively and receives Nash bargaining weight $1 - \alpha$.

Consider first the S2 bargain. Fix a structure $\omega = (F, L, P)$ with $L \geq 1$ and S1 contract set at date t_{s1} . The model chooses the S2 settlement terms according to

$$(t_{s2}^*, C_{s2}^*)(t_{s1}, \omega) = \arg \max_{t_{s2}, C} [\text{GFT}_{s2}^B(t_{s2}, C)]^\alpha [\text{GFT}_{s2}^\ell(t_{s2}, C)]^{1-\alpha},$$

where GFT_{s2}^ℓ is the gain from trade of one representative LTF. Because the L LTFs are symmetric and receive identical settlement terms, their aggregate gain from trade is $L \text{GFT}_{s2}^\ell(t_{s2}, C)$. A collective formulation of the stage-2 bargain would therefore maximize $[\text{GFT}_{s2}^B(t_{s2}, C)]^\alpha [L \text{GFT}_{s2}^\ell(t_{s2}, C)]^{1-\alpha}$. For every candidate pair (t_{s2}, C) :

$$[\text{GFT}_{s2}^B(t_{s2}, C)]^\alpha [L \text{GFT}_{s2}^\ell(t_{s2}, C)]^{1-\alpha} = L^{1-\alpha} [\text{GFT}_{s2}^B(t_{s2}, C)]^\alpha [\text{GFT}_{s2}^\ell(t_{s2}, C)]^{1-\alpha}.$$

Conditional on ω , the number of LTFs L is fixed. $L^{1-\alpha}$ is a strictly positive constant that does not change with cash and date, so multiplying the Nash product by this constant changes its level but not its argmax. The representative-LTF and collective-LTF formulations produce the same (t_{s2}^*, C_{s2}^*) .

The same argument applies to the S1 bargain. At a fixed structure ω , there are F symmetric FTFs. The implemented objective is $[\text{GFT}_{s1}^B(t_{s1}, C)]^\alpha [\text{GFT}_{s1}^f(t_{s1}, C)]^{1-\alpha}$ where the notation suppresses the equilibrium LTF response $\rho(t_{s1}, \omega)$. A collective-FTF formulation would instead use $[\text{GFT}_{s1}^B(t_{s1}, C)]^\alpha [F \text{GFT}_{s1}^f(t_{s1}, C)]^{1-\alpha}$. For every candidate (t_{s1}, C) , the latter equals $F^{1-\alpha} [\text{GFT}_{s1}^B(t_{s1}, C)]^\alpha [\text{GFT}_{s1}^f(t_{s1}, C)]^{1-\alpha}$.

The inclusion of antitrust scrutiny Ψ_{s1} and Ψ_{s2} does not invalidate the above argument. They influence the argmax by entering the corresponding gains-from-trade terms. While the presence of antitrust scrutiny changes the equilibrium settlement terms, it changes them in exactly the same way under the representative-generic and collective-generic formulations. Conditional on ω , the identities $[F \text{GFT}_{s1}^f]^{1-\alpha} = F^{1-\alpha} [\text{GFT}_{s1}^f]^{1-\alpha}$ and $[L \text{GFT}_{s2}^\ell]^{1-\alpha} = L^{1-\alpha} [\text{GFT}_{s2}^\ell]^{1-\alpha}$ remain the same.

E Profit calibration and price prediction

I use regressions of revenue and prices on competition from [Alam and Conti \(2026\)](#) to calibrate per-period profits for brands and generics (Table 6). They also find the marginal cost to be approximately 0.45 of observed mean prices in a market. I set “qtrs since LOE” to 0 for calibration.

[Alam and Conti \(2026\)](#) uses IQVIA’s National Sales Perspective data for the regression. Note that I do not have access to or use that dataset for this current project.

Table 6: Regression showing competitive effects on market shares, revenue, and prices.

	log(share)	log(price)	log(revenue)
Brand	1.590*** (0.080)	0.291*** (0.029)	1.938*** (0.080)
Brand x qtrs since LOE	-0.057*** (0.003)	0.007*** (0.001)	-0.059*** (0.003)
Authorized generic $\times$ Count of generics	-0.133*** (0.013)	-0.143*** (0.005)	-0.278*** (0.013)
Generic $\times$ Count of generics	-0.216*** (0.008)	-0.138*** (0.003)	-0.358*** (0.008)
Brand $\times$ $\mathbb{1}(\text{count of generics} \leq 2)$ x Count of generics	-0.569*** (0.038)	0.038** (0.014)	-0.502*** (0.038)
Brand $\times$ $\mathbb{1}(\text{count of generics} > 2)$ x (Count of generics - 2)	-0.103*** (0.015)	0.100*** (0.006)	-0.007 (0.015)
Generic $\times$ AG presence	-0.300*** (0.057)	-0.182*** (0.021)	-0.475*** (0.057)
Brand $\times$ AG presence	-0.483*** (0.083)	0.004 (0.031)	-0.497*** (0.083)
Molecule-Formulation Fixed Effects	Yes	Yes	Yes
Obs.	30,822	30,822	30,822

Notes: These results are for non-injectable molecule-formulations. The variable “qtrs since LOE” is the number of quarters since loss-of-exclusivity.